Proactive Transfer Admission Control for Emergency Departments

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Problem Definition: Emergency Department (ED) crowding remains a persistent challenge, undermining timely care, elevating clinical risk, and placing substantial strain on hospital resources. While hospitals cannot easily control the arrival of walk-in emergency patients, they often have some flexibility in managing transfer admissions from other hospitals. Temporarily restricting such transfers during periods of high congestion can provide an effective operational lever to alleviate crowding. In this work, we investigate how predictive information, namely forecasts of future demand and inpatient admission delays, can be systematically incorporated into ED admission control decisions. **Methodology/Results:** We develop a queueing model that captures key patient flow dynamics, accounting for a time-varying operating environment, two distinct sources of ED arrivals, and delayed inpatient admissions (i.e., ED boarding). Building on this model, we derive the optimal admission control policy under a fluid approximation. The fluid-based policy can be seamlessly adapted to the stochastic system. It explicitly leverages predictive information, offers clear interpretability and practical implementability, and achieves very good performance. Through extensive numerical experiments and a case study based on real hospital data, we show that by proactively regulating the acceptance of transfer patients based on real-time congestion levels and anticipated future conditions, the proposed approach achieves a superior efficiency frontier balancing ED congestion and transfer acceptance than reactive admission policies. **Managerial Implications:** Our approach provides a systematic framework of integrating predictive analytics to proactively control admission that jointly addresses ED crowding and patient access.

Key words: emergency department, admission control, demand surge, queueing theory, asymptotic analysis

1. Introduction

Emergency department (ED) crowding has emerged as one of the most pervasive and persistent challenges in hospital care delivery. It reflects not only localized operational bottlenecks but also broader systemic strain, and has been shown to adversely affect patient outcomes and contribute to provider burnout (Hoot and Aronsky 2008, Sun et al. 2013). Despite widespread recognition of the problem, hospitals face substantial constraints in expanding capacity. Staffing shortages, rising costs, and the limited availability of physical space make it increasingly difficult for hospitals to simply "scale up" their way out of crowding (Janke and Venkatesh 2025).

Importantly, ED crowding is not merely an ED-level operational issue; it reflects a system-wide patient flow problem. A substantial portion of ED congestion arises from downstream bottlenecks, particularly delays in admitting patients to inpatient units (McKenna et al. 2019). When inpatient beds are unavailable or admission processing is slow, admitted patients remain boarded in the ED, consuming space and staff resources and exacerbating crowding. Therefore, any meaningful approach to ED admission control must account not only for current ED conditions but also for inpatient-side congestion and the anticipated delays in admitting patients.

Given these barriers, there is growing interest in operational strategies that manage, not just accommodate, demand. While hospitals have little control over unscheduled "walk-in" emergency arrivals, many have some flexibility in how they manage inbound inter-facility transfer requests from one ED to another ED. Such ED-to-ED transfers play an important role in regionalized care systems, ensuring that patients receive the appropriate level of care and clinical services (Westfall 2024, Greenwood-Ericksen et al. 2025). Although the national percentage of ED patients in the United States who arrived from another ED is low (estimated at 1.1% in 2020-2022), it has increased over time (Peters et al. 2026) and can be substantially higher at some academic medical centers or referral centers. For example, Moore et al. (2023) study an ED where 13.6% of patients were received from another ED. Moreover, these encounter-based proportions do not capture transfer requests that are declined and therefore do not result in an ED visit. Transfer decisions are subject to applicable requirements, including those of the Emergency Medical Treatment and Labor Act (EMTALA) in the United States (Centers for Medicare and Medicaid Services 2026). Our focus is on academic medical centers and referral hospitals that receive transfer requests from community hospitals seeking care unavailable locally. For these receiving institutions, ED-to-ED transfers constitute a potentially modifiable stream of incoming patients whose timing and volume can meaningfully influence ED congestion (Greenwood-Ericksen et al. 2025). Temporarily regulating such admissions during periods of high crowding offers a promising lever for improving patient flow. In this work, we examine how to design effective admission control policies for ED-to-ED transfer patients (hereafter, transfer patients) to help alleviate ED crowding.

At the same time, recent advances in predictive analytics, combined with the growing availability of granular operational data, have created new opportunities for hospitals to anticipate changes in congestion rather than respond only to current conditions. In particular, forecasts of future ED demand and inpatient admission delays can provide information about how the operating environment is likely to evolve over the coming hours. This raises a natural question: how should such forecasts of time-varying system primitives be translated into transfer-acceptance decisions? In this work, we address this question by developing a forecast-informed admission-control framework that uses anticipated future demand and inpatient processing conditions to assess the congestion consequences of accepting transfer patients.

To study this problem, we develop a queueing model that captures the essential dynamics of ED patient flow relevant to transfer admission control. Our model distinguishes between two streams of arrivals: (1) *Regular Arrivals*, including walk-ins, ambulance arrivals, and referrals from external providers; and (2) *Transfer Requests*, representing patients seeking transfer from other EDs. Admission control is applied exclusively to transfer requests. The model also explicitly captures delays in inpatient admission through a boarding stage, reflecting limited downstream capacity and other operational constraints. By accounting for the time-varying nature of ED operations and these key system features, the model provides a flexible framework for analyzing policies that balance patient waiting costs with the blocking costs associated with rejecting transfer requests.

For analytical development, we first study a deterministic fluid approximation in which the future trajectories of the system primitives are treated as given. This reference problem allows us to characterize how anticipated changes in demand, inpatient admission delays (inpatient processing), and staffing affect transfer-admission decisions. We derive the optimal fluid control and show that these future operating conditions enter the decision through a hitting-time-based congestion metric that summarizes the predicted duration of congestion. This structural characterization leads naturally to a forecast-informed policy for the stochastic system: at each review epoch, unknown future arrival and inpatient-processing rates are replaced by point forecasts constructed from currently available information. The resulting policy retains a simple threshold structure and is easy to implement. Under suitable regularity conditions, we further establish its asymptotic optimality in the many-server limit.

We then evaluate the policy through extensive simulation experiments and a case study based on real hospital data. The synthetic experiments consider different demand-surge patterns, review frequencies, and controlled forecast-error stress tests that vary the magnitude, persistence, horizon dependence, bias, and timing of forecast errors. These experiments allow us to identify which dimensions of forecast quality are most consequential for admission-control performance, highlighting the particular importance of accurately capturing the magnitude and timing of impending congestion. We then provide an end-to-end empirical implementation using data-driven models fitted to real hospital data to forecast ED arrival and inpatient processing rates. The results show that the forecast-informed policy achieves a more favorable trade-off between patient waiting and transfer rejection than commonly used reactive benchmarks.

Taken together, our work makes two primary contributions. First, we provide a structural analysis of transfer admission control in a nonstationary ED with downstream boarding. Using a fluid approximation, we show that the effects of anticipated future demand, inpatient processing, and staffing can be summarized through a congestion-clearing-time metric that determines the optimal admission threshold. While the resulting threshold structure is consistent with classical

admission-control insights, our analysis identifies the underlying congestion externality generated by an additional transfer patients and establishes conditions under which the fluid-based policy is optimal. We further provide an asymptotic optimality guarantee for its implementation in the stochastic system, thereby offering a principled foundation for translating forecasts of future operating conditions into admission decisions.

Secondly, we provide a comprehensive numerical and empirical evaluation of the proposed framework. The synthetic experiments isolate the effects of different forms of forecast error, while the hospital case study implements the policy using forecasting models trained and evaluated on real operational data. Across a broad range of operating conditions, the forecast-informed policy achieves a more favorable congestion-access trade-off than reactive admission rules. Together, these analyses demonstrate both the operational value of incorporating forecasts of future system conditions into transfer admission control and the importance of accurately capturing the magnitude and timing of impending congestion.

1.1. Literature review

Our work is mainly related to two lines of literature.

Admission control in queues Admission control has long been studied in queueing theory as a mechanism for mitigating congestion by balancing waiting costs against blocking or rejection penalties (see, e.g., Heyman 1968, Lewis 2001, Altman et al. 2002, Ormeci 2004, Koçağa and Ward 2010). In many classical stationary settings, the optimal admission policy exhibits a threshold structure (Naor 1969, Stidham Jr 2002, Ward and Kumar 2008). Admission control has also been studied in queueing networks with multiple stages or resources. For example, Silva et al. (2013) analyze admission control in a tandem loss system, although their model is time homogeneous.

Our theoretical model is more closely related to a separate literature on control of nonstationary queueing systems, in which time-varying arrival, service, or capacity rates are treated as primitives of the control problem. Bassamboo et al. (2005) study dynamic routing and admission control in a multiclass many-server system with temporally and stochastically varying arrival rates. Using a multi-scale fluid approximation, they derive asymptotically optimal controls based on the prevailing arrival-rate environment. Čudina and Ramanan (2011) develop a general framework for asymptotically optimal control of time-inhomogeneous queueing networks by first solving an associated fluid control problem. Zayas-Cabán and Lewis (2020) study admission control in loss systems with periodically varying parameters and characterize how optimal admission decisions adjust over predictable demand cycles. These studies establish an important precedent for treating temporal variation in system primitives explicitly and for using fluid or other approximations to obtain tractable control policies. In our reference model, the future trajectories of the primitive

rates are treated as given, and the fluid analysis characterizes how their temporal evolution affects the optimal admission decision. The ED setting, however, introduces an additional source of time variation through downstream inpatient processing. In particular, accepting a transfer patient affects congestion both directly (through the current ED workload) and indirectly (through future boarding dynamics). Our analysis shows that the combined effect of anticipated arrivals, inpatient processing, and staffing can be summarized through a congestion-clearing-time statistic that determines the admission threshold.

A distinct literature studies queueing control under richer forms of information about future stochastic events. Spencer et al. (2014) consider admission control when future arrival and service-event realizations are revealed over a look-ahead window. Motivated by ED diversion, Xu and Chan (2016) study proactive policies that exploit predictions of future arrival and service-token processes and analyze the effect of prediction noise. More recently, Peng (2024) develops a general forecast-information framework for admission control that encompasses a range of information structures, including no future information and full future information. Our information structure is deliberately coarser. We do not assume that the manager observes or predicts future sample-path realizations of individual arrivals, service completions, or inpatient-bed releases. Instead, with a view towards practical implementation, we replace the future primitive-rate trajectories in the reference model with point forecasts of aggregate arrival and inpatient-processing rates available at each decision epoch.

Relatedly, Bertsimas et al. (2022) operationalize machine-learning forecasts by embedding them into hospital information systems to support daily bed-placement decisions. More broadly, (Ata and Peng 2020, Delana et al. 2021, Ao et al. 2024) study how anticipatory information can be used to improve resource-allocation and service decisions in stochastic systems.

Managing Congestion in the ED The conceptual model of Asplin et al. (2003) frames ED crowding in three interdependent components: input, throughput, and output. Crowding is essentially an imbalance where input exceeds throughput and output capacity in the acute care system. Extensive work in the emergency medicine literature has documented the causes and consequences of ED crowding, including its effects on patient safety, service delays, and provider well-being (see, for example, Hoot and Aronsky 2008, Bernstein et al. 2009, Morley et al. 2018). Practices in ED-to-ED transfers and acceptance decisions have also been studied (Greenwood-Ericksen et al. 2025).

Within operations research, ED congestion has been studied from multiple angles using stochastic modeling, empirical analysis, and simulation. Queueing models have been widely applied to support operational decisions. For example, Green et al. (2006) and Yom-Tov and Mandelbaum (2014) use queueing analysis to determine appropriate staffing levels under varying demand conditions. Chan et al. (2022) study a fluid control problem for nurse reassignment in an ED. Jacobson et al.

(2012) and Saghafian et al. (2012) show that streaming and prioritization schemes can be effective tools for improving system performance, particularly when the ED experiences extreme levels of crowding. Shi et al. (2016) develop a stochastic network model of inpatient operations and show that inpatient discharge policies can have a substantial impact on ED boarding times. A parallel empirical literature investigates how ED clinicians adapt their behavior in response to congestion. For example, Song et al. (2015) analyze how alternative queue configurations influence ED physician productivity. Batt and Terwiesch (2017) document the phenomenon of early task initiation, where upstream staff (e.g., triage nurses) begin tasks typically performed by downstream providers when the ED becomes busy. Ding et al. (2019) and Li et al. (2021) show that prioritization decisions also shift with crowding. Adepoju et al. (2023) studied how reactive hospital-wide responses to ED crowding may have unintended negative consequences. Recently, Canellas et al. (2025) consider fair patient prioritization and placement in ED operations, using a predictive-prescriptive mixed integer linear program that breaks predicted ED length of stay into actionable components. Our work contributes to this literature by examining a complementary and understudied operational lever: admission control for inter-facility transfer patients.

1.2. Notations

For integer $n \geq 1$, we denote $[n] = \{1, 2, \ldots, n\}$ as the set of integers from 1 to n. For $x \in \mathbb{R}$, denote $x_+ = \max\{x, 0\}$. For set S, denote $|S|$ as its cardinality.

2. Problem Formulation

We model the emergency department (ED) as a multi-server queue with a potentially time-varying number of servers, denoted by $c(t)$. Patients arrive through two distinct streams: (1) External arrivals are patients entering the ED from outside the hospital system, such as walk-ins, ambulance arrivals (not from another ED), and referrals from external healthcare providers; and (2) Transfer requests are patients seeking transfer from other EDs. External arrivals follow a time-varying Poisson process with rate $\lambda_1(t)$, while transfer requests arrive according to a Poisson process with rate $\lambda_2(t)$. We denote the total arrival rate by $\lambda(t) = \lambda_1(t) + \lambda_2(t)$. Patients are served on a first-come-first-served basis, with service times that are exponentially distributed with rate μ. While waiting for service, patients may abandon the queue if their patience expires; patience times are assumed to be exponentially distributed with rate $\theta < \mu$. Upon completing ED treatment, a patient either leaves the hospital directly with probability p or is admitted to an inpatient ward (IW) with probability $1 - p$. Patients requiring inpatient admission experience a boarding delay prior to transfer, reflecting limited bed availability or administrative bottlenecks on the inpatient side. Each boarding patient completes boarding at a time-varying instantaneous rate $\nu(t)$, which we refer to as the processing rate. Importantly, patients remain in an ED service position while boarding

and therefore continue to occupy ED server capacity until their transfer to the inpatient ward is completed.

We consider an admission control policy under which the hospital determines whether to accept incoming transfer requests. Specifically, the control variable $G(t) \in \{0,1\}$ indicates whether transfer requests arriving at time t are accepted, $G(t)=1$, or rejected, $G(t)=0$.

Let $X(t)$ denote the total number of patients in the ED who have not yet completed service (treatment), including both those waiting in the queue and those currently receiving service. In addition, let $Z(t)$ denote the number of patients currently in service and $Q(t)$ denote the number of patients waiting in the queue. Then, $X(t)=Q(t)+Z(t)$. We further let $B(t)$ denote the number of patients boarding in the ED while awaiting transfer to the IWs.

Let A_1, A_2, S_d, S_b, D, and R be independent rate-1 Poisson processes representing external arrivals, transfer requests, service completions, patient abandonments, and inpatient transfer completions, respectively. The system dynamics can then be characterized by

$$\begin{aligned} X(t) =& X(0) + A_1\left(\int_0^t \lambda_1(t)dt\right) + A_2\left(\int_0^t \lambda_2(t)G(t)dt\right) - S_d\left(p\int_0^t \mu Z(t)dt\right) \\ & - S_b\left((1-p)\int_0^t \mu Z(t)dt\right) - D\left(\int_0^t \theta Q(t)dt\right), \\ B(t) =& B(0) + S_b\left(\int_0^t (1-p)\mu Z(t)dt\right) - R\left(\int_0^t \nu(t)B(t)dt\right), \end{aligned} \tag{1}$$

Without loss of optimality, we assume the system is work-conserving, so that

$$Z(t) = \min\{c(t) - B(t),\, X(t)\}.$$

An admission control policy $G(t)$ may depend on the current system state $(X(t), B(t))$ and on information available at time t about the future operating environment. In particular, let $\widehat{\lambda}_i(s \mid t)$, $i \in 1,2$, and $\widehat{\nu}(s \mid t)$, $s \geq t$, denote forecasts of future arrival and inpatient processing rates constructed using information available at time t. We assume these forecasts are available to support admission decisions. We also assume that future staffing levels $\{c(s), s \geq t\}$ are known, reflecting the fact that staffing schedules are typically determined in advance. Importantly, these forecasts describe the anticipated evolution of the system primitives; they do not reveal future sample-path realizations of individual arrivals, service completions, or inpatient transfers.

Let π denote an admissible admission control policy, and use superscripts to indicate dependence on π, e.g., $X^\pi(t)$, $B^\pi(t)$. We associate a holding cost h per patient per unit time spent waiting in the queue, and a blocking cost l for each rejected transfer request. The objective is to minimize the expected total cost over a planning horizon T:

$$V^\pi(x,b) = \mathbb{E}\left[\int_0^T h(X^\pi(t) - Z^\pi(t))dt + lL^\pi(T) \,\middle|\, X(0)=x, B(0)=b\right] \tag{2}$$

where $L^\pi(T)$ denotes the number of transfer requests rejected by time T. Note that

$$L^\pi(T) \sim \tilde{E}\left(\int_0^T \lambda_2(t)\left(1 - G^\pi(t)\right)dt\right) \tag{3}$$

where $\tilde{E}$ is a rate-1 Poisson process. The planning horizon T is chosen to be sufficiently long for the system to fully recover from any demand surge; a formal definition is provided in Appendix B.

The model deliberately abstracts from several operational details of ED patient flow, such as unit-level bed heterogeneity, patient-level acuity, and service prioritization. Our objective is not to reproduce the full complexity of ED operations but to isolate the central trade-off in transfer admission control: accepting a transfer increases future congestion, whereas rejecting it incurs an immediate blocking cost. This trade-off is primarily driven by the interaction among the current ED and boarding census, anticipated future demand, and expected inpatient processing conditions. This level of aggregation is also consistent with practice, where transfer-admission decisions are typically based on system-level indicators such as ED census, boarding levels, and expected discharges rather than detailed patient- or unit-level information. The model's state variables therefore align with both the information routinely available to decision makers and the organizational level at which transfer-admission decisions are implemented.

3. Fluid Analysis

For analytical development, we first consider a deterministic reference model in which the future primitive trajectories $\{\lambda_1(s), \lambda_2(s), \nu(s), c(s) : s \geq t\}$ are treated as given. This reference model allows us to characterize how anticipated future operating conditions should enter the admission decision. Even under this reference setting, the stochastic control problem in (2) remains analytically intractable because of (i) the nonstationary primitives $(\lambda_1(t), \lambda_2(t), \nu(t), c(t))$ and (ii) the complexity induced by the joint evolution of the ED census and boarding population. Moreover, even when numerical solutions can be obtained for stylized instances, they provide limited structural insight into how forecasted operating conditions should influence admission decisions.

To obtain a tractable characterization of the optimal policy, we therefore study a deterministic fluid approximation. The fluid model captures the mean dynamics of the underlying queueing system and is particularly well suited to the congested, time-varying regimes of interest. Importantly, the fluid analysis allows us to identify a low-dimensional summary of future operating conditions that governs the admission decision, which will later serve as the basis for the forecast-based policy implemented in the stochastic system.

Let $x(t)$, $z(t)$, $b(t)$, and $g(t)$ denote the fluid analogues of $X(t)$, $Z(t)$, $B(t)$, and $G(t)$, respectively. The fluid dynamics satisfy

$$\begin{aligned} \dot{x}(t) &= \lambda_1(t) + g(t)\lambda_2(t) - \mu z(t) - \theta\big(x(t) - z(t)\big), \\ \dot{b}(t) &= (1-p)\mu z(t) - \nu(t) b(t), \end{aligned} \tag{4}$$

where $z(t) = \min\{c(t) - b(t),\, x(t)\}$ and $\dot{x}(t) = dx(t)/dt$, $\dot{b}(t) = db(t)/dt$.

3.1. Single Congestion Shock

We first analyze a setting with a single congestion shock beginning at time 0 and ending at time $\kappa > 0$. Such a shock may be driven by a demand surge, a temporary reduction in the processing rate, or both. We impose the following regularity and load conditions.

ASSUMPTION 1. *1. The functions $\lambda_1(t)$, $\lambda_2(t)$, and $\nu(t)$ are piecewise monotone with a finite number of pieces, and $c(t) \equiv c$.*

2. The initial condition satisfies $x(0) + b(0) \geq c(0)$. There exists $\epsilon_1 > 0$ such that for all $t \in [0, \kappa)$,

$$\lambda_1(t) - \max\{p\mu c(t),\, \nu(t)c(t)\} \geq \epsilon_1.$$

3. There exists $\epsilon_2 > 0$ such that for all $t \geq \kappa$,

$$\lambda(t) - \min\{p\mu c(t),\, \nu(t)c(t)\} \leq -\epsilon_2,$$

where $\lambda(t) = \lambda_1(t) + \lambda_2(t)$.

Assumption 1.1 is a standard regularity condition ensuring existence and uniqueness of solutions to the fluid ODEs and enabling convergence arguments under many-server scaling. Assumptions 1.2–3 impose a pronounced overload prior to time κ and a subcritical regime thereafter, which together ensure that the fluid queue remains positive before κ and is eventually cleared after κ. While these conditions can be relaxed, the essential requirement for our analysis is that the fluid queue does not hit zero during the overload phase and eventually drains to zero in finite time.

Consistent with the deterministic reference model, we assume when choosing $g(t)$, the decision maker treats the future primitive trajectories $\{\lambda_1(s), \lambda_2(s), \nu(s), c(s) : s \geq t\}$ as given.

Let $q(t) = x(t) - z(t)$ denote the fluid queue length. Define the (fluid) queue-clearing time

$$\tau \;=\; \inf\{t \geq \kappa :\; x(t) + b(t) = c(t)\}.$$

Under Assumption 1, $\tau < \infty$ and $q(t) = 0$ for all $t \geq \tau$. The fluid optimal control problem corresponding to (2) can be written as

$$\begin{aligned} \min_{g(\cdot)} \quad & \int_0^\tau \Big[h\big(x(t) - z(t)\big) + l\lambda_2(t)\big(1 - g(t)\big) \Big]\, dt \\ \text{s.t.} \quad & (4), \qquad z(t) = \min\{x(t), c(t) - b(t)\}, \qquad 0 \leq g(t) \leq 1. \end{aligned} \tag{5}$$

For $t \geq 0$, define the hitting time $H_t(x, b)$ as the time required for the system to return to a zero-queue state when *all* transfers are accepted (i.e., $g(\cdot) \equiv 1$), starting from state (x, b) at time t:

$$H_t(x, b) := \inf\Big\{ \Delta > 0 :\; \tilde{x}(t + \Delta) + \tilde{b}(t + \Delta) \leq c(t + \Delta) \;\Big|\; \tilde{x}(t) = x,\; \tilde{b}(t) = b \Big\}, \tag{6}$$

where $(\tilde{x}(\cdot), \tilde{b}(\cdot))$ is the solution to the uncontrolled ODE system

$$\begin{aligned} &\tilde{x}(t) = x, \qquad \tilde{b}(t) = b, \\ &\dot{\tilde{x}}(u) = \lambda(u) - \mu\tilde{z}(u) - \theta\,\tilde{q}(u), \qquad u \geq t, \\ &\dot{\tilde{b}}(u) = (1 - p)\mu\tilde{z}(u) - \nu(u)\tilde{b}(u), \qquad u \geq t, \end{aligned}$$

with $\tilde{z}_(u) = \min\{\tilde{x}(u), c(u) - \tilde{b}(u)\}$ and $\tilde{q}(u) = \tilde{x}(u) - \min\{\tilde{x}(u), c(u) - \tilde{b}(u)\}$. For boundary states satisfying $x + b = c(t)$, we define $H_t(x, b)$ by continuity as $H_t(x, b) = \lim_{x' \downarrow (c(t)-b)} H_t(x', b)$. Intuitively, $H_t(x, b)$ measures the duration of congestion implied by the current state and the future primitives $(\lambda(\cdot), \nu(\cdot), c(\cdot))$.

The following theorem characterizes the optimal admission control for (5).

THEOREM 1. *Under Assumption 1, an optimal control for* (5) *is given by*

$$g^*(t) = \mathbf{1}\left\{\frac{h}{\theta}\Big(1 - \exp(-\theta H_t(x(t), b(t)))\Big) \le l\right\}. \tag{7}$$

The optimal policy balances the immediate blocking cost l against the marginal system-wide waiting cost generated by an additional admission. In the single-shock setting, this marginal cost declines as the optimal trajectory approaches recovery, yielding a single-switch structure: the policy either accepts all transfers from the outset or rejects them initially and then switches to acceptance. After the switch, all subsequent transfers are accepted, so the remaining congestion duration coincides with the all-accept hitting time H_t. Before the switch, the all-accept trajectory cannot clear sooner than the optimal trajectory, so the hitting-time comparison also prescribes rejection. The resulting threshold rule therefore bases admission decisions on the duration of congestion implied by the current state and future operating conditions.

Let $f(H) := \frac{h}{\theta}\big(1 - e^{-\theta H}\big)$. To interpret the congestion-cost index $f(H_t)$, consider the all-accept fluid trajectory starting from the current state, and increase the initial treatment population by a small amount ε, leaving the boarding population unchanged. While both trajectories remain congested, their boarding dynamics and service allocations coincide, and the difference in their queue lengths satisfies $\Delta q(t + s) = \varepsilon e^{-\theta s}$, for $0 \le s < H_t$. Integrating this marginal queue perturbation over the reference congestion episode gives the cumulative holding cost per unit of additional fluid

$$h \int_0^{H_t} e^{-\theta s}\, ds = \frac{h}{\theta}\left(1 - e^{-\theta H_t}\right) = f(H_t).$$

This quantity captures waiting-cost effects across the system. An admitted patient may begin treatment before congestion clears, while its effect on the queue persists through delays imposed on other patients.

Two limiting cases further clarify the rule. If $h/\theta \le l$, then $f(H_t) \le l$ for every congestion horizon, so all transfers are accepted. In the absence of abandonment, taking $\theta \downarrow 0$ gives $f(H) = hH$, and the acceptance condition becomes $hH_t \le l$.

The detailed proof of Theorem 1 is provided in Appendix A.1, where we apply Pontryagin's Minimum Principle and verify the sufficient conditions for optimality.

3.2. Multiple Congestion Shocks

The single-shock setting in Section 3.1 yields a particularly transparent characterization because the system experiences only one congestion episode before fully clearing. In practice, however, the ED may encounter multiple congestion episodes over the planning horizon due to repeated demand surges or recurrent reductions in inpatient processing rate. In this subsection, we extend the analysis to this more general setting. Our goal is twofold. First, we show that the optimal admission policy in the multi-surge setting still has a threshold structure. Second, we explain why the exact threshold statistic is substantially more complicated than in the single-surge case, and motivate a simpler approximation based on the single-surge policy.

To formulate the multi-surge setting, we impose a piecewise-overload structure analogous to Assumption 1.

ASSUMPTION 2 **(Piecewise overload/underload structure)**. *The functions $\lambda_1(t)$, $\lambda_2(t)$, and $\nu(t)$ are piecewise monotone with a finite number of pieces, and $c(t) \equiv c$. Moreover, there exist times $0 = \kappa_0 < \kappa_1 < \cdots < \kappa_M < \infty$ such that on each interval $[\kappa_{m-1}, \kappa_m)$, the net input condition is of a fixed sign; that is, the system is either overloaded or underloaded throughout the interval, i.e., either*

$$\lambda_1(t) - \max\{p\mu c(t),\, \nu(t)c(t)\} \geq \epsilon_m^+$$

for some $\epsilon_m^+ > 0$, or

$$\lambda(t) - \min\{p\mu c(t),\, \nu(t)c(t)\} \leq -\epsilon_m^-$$

for some $\epsilon_m^- > 0$, where $\lambda(t) := \lambda_1(t) + \lambda_2(t)$. For $t \geq \kappa_M$

$$\lambda(t) - \min\{p\mu c, \nu(t)c\} \leq -\epsilon_{M+1},$$

for some $\epsilon_{M+1} > 0$.

A key distinction from the single-surge case is that the timing of congestion is endogenous. Even if the instantaneous load becomes unfavorable at some time, the queue may take time to build up before becoming positive; similarly, once the load turns favorable again, the queue may persist for some time before it fully dissipates. It is therefore useful to distinguish between primitive stress intervals, defined from the model inputs, and the resulting *congestion periods*, defined from the fluid state. Recall that the fluid queue length is

$$q(t) = x(t) - z(t) = \big(x(t) + b(t) - c(t)\big)^+.$$

Let $(x(\cdot), b(\cdot))$ be the fluid trajectory induced by an admissible control $g(\cdot)$. A *congestion period* is a maximal interval $I_k = [\sigma_k, \tau_k]$ such that $q(t) > 0$ for $t \in (\sigma_k, \tau_k)$. An *uncongestion period* is a maximal interval $G_k = [\tau_k, \sigma_{k+1}]$ such that $q(t) = 0$ for $t \in G_k$. Thus, a congestion period is an interval during which a positive queue is present, whereas an uncongestion period is an interval during which the queue has fully cleared.

3.2.1. Two-Shock Case We first consider the simplest multi-shock environment in which there are two congestion shocks. Let $I_1 = [0, \tau_1]$, i.e., $\sigma_1 = 0$, and $I_2 = [\sigma_2, \tau_2]$ denote the two congestion periods, separated by the uncongested gap $G_1 = [\tau_1, \sigma_2]$. Note that if no such uncongested gap exists, then the two shocks effectively merge into a single congestion episode, reducing the problem to the single-shock case.

Let $p_x(t)$ and $p_b(t)$ denote the adjoint processes associated with the state variables $x(t)$ and $b(t)$ along an optimal trajectory, as given by Pontryagin's Maximum Principle; see Appendix A for the full adjoint system and transversality conditions. These adjoint variables encode the marginal value of perturbations to the fluid state: $p_x(t)$ is the marginal effect on the objective of an additional ED arrival at time t, while $p_b(t)$ is the marginal effect of an additional boarding patient at time t. As in the single-shock case, the Hamiltonian is affine in the control variable $g(t)$; see Appendix A. Therefore, the optimal control has threshold form: $g^*(t) = \mathbf{1}\{p_x(t) \le l\}$. It is convenient to define the *effective congestion horizon*

$$H_t^e := f^{-1}(p_x(t)), \text{ where } f(H) := \frac{h}{\theta}\big(1 - e^{-\theta H}\big).$$

Then, the optimal policy can be written equivalently as $g^*(t) = \mathbf{1}\{f(H_t^e) \le l\}$.

THEOREM 2 **(Optimal policy in the two-surge case)**. *In the two-surge setting described above, the optimal admission policy is given by*

$$g^*(t) = \mathbf{1}\{p_x(t) \le l\} = \mathbf{1}\{f(H_t^e) \le l\}, \qquad t \ge 0.$$

The effective congestion horizon H_t^e admits the following characterization:
(i) ***Congestion periods.*** *For $t \in I_k = [\sigma_k, \tau_k]$, $k = 1, 2$,*

$$H_t^e = (\tau_k - t) + C_k, \qquad C_k := H_{\tau_k}^e.$$

In particular, $C_2 = 0$, while C_1 captures the impact of the second congestion episode on the first.
(ii) ***Uncongested period.*** *For $t \in G_1 = [\tau_1, \sigma_2]$, the adjoint variables satisfy*

$$p_b(t) = \exp\left(-\int_t^{\sigma_2} \nu(r)\, dr\right) p_b(\sigma_2),$$

and

$$p_x(t) = e^{-\mu(\sigma_2 - t)} p_x(\sigma_2) + (1-p)\mu \int_t^{\sigma_2} e^{-\mu(u-t)} \exp\left(-\int_u^{\sigma_2} \nu(r)\, dr\right) du\, p_b(\sigma_2).$$

Theorem 2 shows that future congestion episodes influence current admission decisions through carry-over terms propagated backward across time. This structure extends naturally to a general multi-shock setting. In particular, for $k = 1, \ldots, K$, on a congestion period $I_k = [\sigma_k, \tau_k]$,

$$H_t^e = (\tau_k - t) + C_k, \text{ for } t \in I_k,$$

where $C_k = H_{\tau_k}^e$ summarizes the effect of subsequent congestion episodes. Thus, the effective congestion horizon consists of the remaining duration of the current congestion episode plus a continuation

term capturing future congestion. During an uncongestion period G_k, the queue is empty, but an additional patient present at time $t \in G_k$ may still affect the system when the next congestion episode begins. Let $p_x(\sigma_{k+1})$ and $p_b(\sigma_{k+1})$ denote the backward terminal values at the onset of the next congestion period. Then, for $t \in G_k$,

$$p_b(t) = \exp\left(-\int_t^{\sigma_{k+1}} \nu(r)\,dr\right) p_b(\sigma_{k+1}),$$

and

$$p_x(t) = e^{-\mu(\sigma_{k+1}-t)} p_x(\sigma_{k+1}) + (1-p)\mu \int_t^{\sigma_{k+1}} e^{-\mu(u-t)} \exp\left(-\int_u^{\sigma_{k+1}} \nu(r)\,dr\right) du\, p_b(\sigma_{k+1}). \tag{8}$$

The two terms in (8) capture the two channels through which a patient present at time t can carry congestion effects into the next episode. The first corresponds to the patient remaining in ED treatment until σ_{k+1}. The second corresponds to the patient completing ED treatment at some $u \in [t, \sigma_{k+1}]$, requiring inpatient admission, and remaining boarded in the ED until σ_{k+1}.

Representation (8) also shows that this intertemporal carry-over weakens as the uncongested gap becomes longer. In particular, if $\nu(r) \geq \underline{\nu} > 0$ on G_k, both propagation channels are exponentially attenuated with the distance to the next congestion episode. Consequently, the carry-over term C_k becomes small as $\sigma_{k+1} - \tau_k$ increases. Intuitively, a long uncongested interval gives the system time to absorb the effect of current admissions before the next surge arrives, so sufficiently distant future congestion has only a limited influence on current admission decisions.

3.2.2. Accuracy of the H_t-Based Policy The exact effective horizon H_t^e incorporates the influence of all future congestion episodes and is therefore difficult to compute in real time. For operational use, we consider the simpler H_t-based policy $g^H(t) = \mathbf{1}\{f(H_t) \leq l\}$, where H_t is the all-accept queue-clearing time defined in (6). Unlike H_t^e, which incorporates the carry-over effect of later congestion episodes, H_t summarizes only the congestion implied by the current state and the local all-accept recovery trajectory. We next assess the quality of this approximation in the two-surge setting.

Use a common fixed horizon T extending beyond the second congestion episode, and let

$$J(g) = \int_0^T \left[hq^g(t) + l\lambda_2(t)(1-g(t))\right] dt.$$

under an admissible control $g(\cdot)$. Directly characterizing the exact gap $J(g^H) - J(g^*)$ is difficult because changing the control alters both the state trajectory and the future switching structure of the optimal-control problem. We therefore examine the local consequences of this approximation.

Let (x^*, b^*) denote this trajectory, let g^* be its admission control, and define $\bar{g}^H(t) := g^H(t, x^*(t), b^*(t))$. Thus, $\bar{g}^H$ is the action prescribed by the hitting-time rule at the optimal state.

LEMMA 1 **(First-variation representation)**. *For every admissible control g, the one-sided directional derivative at g^* is*

$$DJ(g^*; g - g^*) = \int_0^T \lambda_2(t)\big(p_x(t) - l\big)\big(g(t) - g^*(t)\big)\,dt,$$

where (p_x, p_b) is the adjoint along the optimal trajectory, with zero terminal values at T.

Define the first-order mismatch penalty

$$\Delta_H^{(1)} := \int_0^{\tau_2} \lambda_2(t)(p_x(t) - l)(\bar{g}^H(t) - g^*(t))\, dt. \tag{9}$$

For $g^\varepsilon = g^* + \varepsilon(\bar{g}^H - g^*)$, the lemma gives $J(g^\varepsilon) - J(g^*) = \varepsilon\Delta_H^{(1)} + o(\varepsilon)$ for $\varepsilon \downarrow 0$.

Let $K := f^{-1}(l)$ denote the critical congestion horizon at which the marginal congestion cost equals the rejection cost. In the single-surge setting, transfers are accepted if and only if the remaining congestion horizon does not exceed K.

PROPOSITION 1 **(A bound on the first-order mismatch penalty)**. *Suppose $\lambda_2(t) \equiv \lambda_2$ and $0 \le C_1 \le K$. Then $\Delta_H^{(1)} = \Delta^{\text{gap}} + \Delta^{\text{carry}}$, where*

$$\Delta^{\text{gap}} = \lambda_2 \int_{\tau_1}^{\sigma_2} (p_x(t) - l)^+\, dt \quad \textit{and} \quad \Delta^{\text{carry}} = \lambda_2 \int_{\mathcal{M}_1} (p_x(t) - l)\, dt,$$

$\mathcal{M}_1 := \{t \in (0, \tau_1) : p_x(t) > l,\ H_t(x^*(t), b^*(t)) \le K\}$. The carry-over term satisfies

$$0 \le \Delta^{\text{carry}} \le \frac{\lambda_2 h}{2} e^{-\theta K} C_1^2.$$

Proposition 1 identifies two distinct sources of approximation error. The first, Δ^{gap}, arises during the uncongested gap: the hitting-time rule accepts transfers, whereas the optimal rule may reject them in anticipation of the second episode. The second, Δ^{carry}, arises during the first congestion period: omitting the continuation effect can also lead the hitting-time rule to accept when rejection is optimal. The proposition also clarifies when these omissions are innocuous. In particular, if $\tau_2 - \sigma_2 \le K$, i.e., the second congestion episode is too short to make proactive rejection worthwhile, then $\Delta^{\text{gap}} = 0$. $\Delta^{\text{carry}} = O(C_1^2)$ as $C_1 \downarrow 0$, which indicates that a longer uncongested gap attenuates the continuation effect on decisions in the first episode.

To assess whether these first-order insights translate into meaningful performance differences in the stochastic system, we further compare the H_t-based rule with the fluid-optimal policy numerically. Both policies are implemented in the two-surge stochastic system described in Section 5, with review interval $\delta = 1$. Figure 1 compares the performance of the two polices. The left panel traces the efficiency frontier between average waiting time and the fraction of rejected transfer requests as l/h varies. The H_t-based frontier closely tracks the fluid-optimal frontier, indicating little loss from replacing H_t^e with the simpler local hitting-time statistic H_t. The right panel reports total cost, with $h = 1$, together with 95% confidence bands based on 200 independent simulation replications. The two policies yield nearly indistinguishable costs over most of the range of l, with a small gap only when rejection is relatively inexpensive and the admission decision is therefore more sensitive to omitted future-congestion effects.

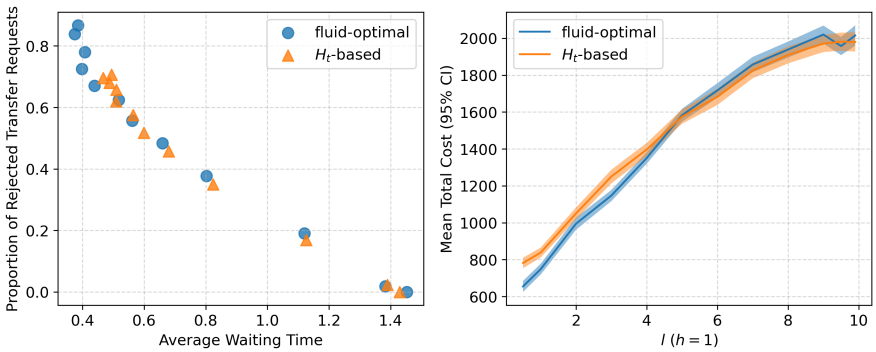


Figure 1 Comparison between H_t-based policy (orange) and fluid-optimal policy (blue).

3.3. Extensions to Patient Heterogeneity

The structural insights of Section 3.1 extend to certain forms of patient heterogeneity. Suppose transfer classes differ only in their rejection costs l_k, while sharing the same service, abandonment, inpatient-admission, and holding-cost parameters. An admitted patient then generates the same marginal congestion cost regardless of class, and the optimal decision compares this common cost with the class-specific rejection cost l_k. Similarly, if patients differ only in ED service priority, work conservation leaves the aggregate fluid dynamics unchanged. Priority determines which patients receive service but does not change the aggregate congestion cost.

Section 6.1 considers an extension motivated by the empirical setting: external and transfer patients have different probabilities of requiring inpatient admission, and transfers receive priority for ED service. These differences affect both the fluid dynamics and the marginal congestion cost of admitting a transfer. We use the heterogeneous fluid model to predict the time until congestion clears and apply the hitting-time rule from Section 3.1 as a heuristic.

Appendix E allows for broader differences between patients. It considers high- and low-priority classes, denoted by H and L, with preemptive priority for class H. External arrivals and transfer requests may have different class compositions, and transfer admission decisions are made separately by class. The classes may differ in service and abandonment rates, probabilities of requiring inpatient admission, and holding and rejection costs. Admitting a high-priority patient can delay lower-priority patients, so a single congestion-clearing time cannot capture every admission consequence. We therefore develop a class-specific look-ahead heuristic that accounts for these cross-class effects: it accepts a transfer when the predicted congestion cost of admission does not exceed the cost of rejection.

4. Admission Control Policies for the Stochastic System

In this section, we describe how the fluid-optimal policy can be implemented in the original stochastic system and discuss several practical considerations that arise in operational settings, including prediction error and limited decision-update frequency. We assume $0 \le l < h/\theta$, so the admission control decision is non-trivial.

4.1. Fluid-Inspired Admission Control Rule

The fluid analysis yields a simple, interpretable admission rule that can be directly applied to the stochastic system. At any time t, the decision maker evaluates the condition

$$\frac{h}{\theta}\left(1-\exp\Big(-\theta H_t\big(X(t),B(t)\big)\Big)\right) > l.$$

If the inequality holds, incoming transfer requests are rejected; otherwise, they are accepted. Here, $(X(t), B(t))$ denote the current stochastic system state, and $H_t(\cdot)$ is the congestion hitting-time metric computed from the fluid model. Operationally, $H_t(X(t), B(t))$ represents the predicted time required for congestion to clear if all transfer requests were accepted from time t onward.

This policy has several appealing features from an implementation standpoint. First, it depends only on low-dimensional, routinely monitored state variables (ED census and boarding population). Second, future arrival rates, processing rates, and staffing levels enter the decision only through the scalar quantity H_t, providing a natural way to aggregate predictive information into a single operational signal.

Although the admission control rule is derived from a deterministic fluid approximation, it admits a rigorous justification for the underlying stochastic system. In particular, under a many-server scaling regime in which arrival rates and capacity grow proportionally, the fluid-translated policy is asymptotically optimal. This result provides theoretical support for using the fluid hitting time as the decision-relevant quantity for admission control in large, highly-utilized EDs. The precise statement of the asymptotic optimality result, along with the technical assumptions and proof, is provided in Appendix B.

4.2. Prediction Error and Imperfect Information

In practice, future arrival and inpatient processing rates are not known exactly and must be forecast using information available at the time of decision. Let $\widehat{\lambda}(s \mid t)$ and $\widehat{\nu}(s \mid t)$, $s \geq t$, denote the forecasts available at review epoch t. We implement the fluid-based policy in a certainty-equivalent manner: conditional on these forecast trajectories, we evaluate the deterministic fluid dynamics and compute the corresponding predicted congestion-clearing time, denoted by $\widehat{H}_t$. The admission rule then rejects transfer requests whenever

$$\frac{h}{\theta}\left(1-\exp\Big(-\theta \widehat{H}_t\big(X(t),B(t)\big)\Big)\right) > l.$$

The resulting decision is implemented until the next review epoch, at which point the system state and forecasts are updated and the hitting time is recomputed. Thus, the policy requires forecasts only of aggregate future arrival and processing rates. Accordingly, the fluid optimality result should be interpreted as conditional on a given trajectory of the system primitives, while Sections 5 and 6 examine the performance implications of replacing these trajectories with imperfect and data-driven forecasts.

When detailed forecasts of the future rate trajectories are unavailable, the same framework can be implemented using coarser estimates of the anticipated operating environment. Let $\bar{\lambda}_t$ and $\bar{\nu}_t$ denote baseline estimates of the future arrival and inpatient processing rates constructed using information available at time t, for example from historical averages for comparable time periods or other simple forecasting models. We then compute an approximate hitting time $\bar{H}_t$ by evaluating the fluid dynamics under constant future rates $\widehat{\lambda}(s \mid t) = \bar{\lambda}_t$ and $\nu(s \mid t) = \bar{\nu}_t$ for $s \geq t$. This provides a simple fallback implementation when detailed time-varying forecasts are unavailable.

Several features of the policy help mitigate the impact of prediction error. First, the current system state $(X(t), B(t))$ provides real-time information about congestion, reducing the policy's reliance on forecasts alone. Second, the hitting-time metric summarizes the predicted arrival and processing rates over the future congestion horizon, so small, transitory forecast errors may have only a limited effect on the resulting estimate. Third, the admission decision depends only on whether the predicted hitting time crosses a fixed threshold. Consequently, moderate forecast errors are unlikely to change the decision when the system is sufficiently far from the switching boundary. When the predicted hitting time is close to the threshold, forecast errors may change the admission decision; however, in this region the marginal congestion cost of accepting a transfer is close to the blocking cost of rejecting it, so the associated performance loss is expected to be small. These observations suggest that the policy should be relatively robust to modest forecast errors, while persistent biases or substantial errors in the magnitude or timing of future congestion may lead to more meaningful performance degradation. We examine these effects numerically in Section 5.2.

4.3. Periodic Review and Implementation Considerations

In operational environments, admission decisions are rarely updated continuously. Moreover, frequent re-evaluation can lead to undesirable decision volatility driven by stochastic fluctuations in the system state, causing the hitting-time metric H_t to oscillate around the decision threshold. To address these practical considerations, we consider a periodic review implementation in which admission decisions are updated at discrete intervals of length δ.

Specifically, at review times $k\delta$, $k = 0, 1, 2, \ldots$, the admission decision is fixed over the interval $[k\delta, (k+1)\delta)$. Transfer requests are rejected during this interval if

$$H_{k\delta}\big(X(k\delta), B(k\delta)\big) - a\delta > -\frac{1}{\theta}\log\left(1 - \frac{l\theta}{h}\right), \tag{10}$$

where $a \geq 0$ is a tuning parameter introduced to compensate for the time lag between reviews.

The review interval δ governs a trade-off between responsiveness and stability. Smaller values of δ allow the policy to react quickly to changes in congestion, but may result in frequent switching driven by noise. Larger values of δ yield more stable decisions and lower administrative burden. Importantly, because the policy depends on whether H_t exceeds a fixed threshold, decision quality

is relatively insensitive to δ when H_t is well above or well below the threshold. When H_t is close to the threshold, the performance difference between accepting and rejecting transfers is small, so less frequent updates can help reduce volatility without materially affecting outcomes. We illustrate these effects through numerical experiments in Sections 5 and 6.

5. Numerical Experiments

In this section, we evaluate the performance of the proposed look-ahead admission control policy through a set of synthetic numerical experiments. Our primary objective is to quantify the operational value of incorporating predictive information into transfer admission decisions, relative to policies that rely solely on contemporaneous congestion measures. Because admission control inherently involves a trade-off between congestion mitigation and access for transfer patients, policy performance cannot be meaningfully evaluated at a single operating point. Accordingly, throughout this section we summarize performance using trade-off curves between the average waiting time and the fraction of rejected transfer requests.

Policy comparison. We compare the proposed look-ahead policy against a class of state-dependent benchmark policies motivated by current hospital practice and by optimal admission control rules in queueing models without predictive information. These benchmark policies rely exclusively on contemporaneous congestion indicators and do not incorporate forecasts of future demand or processing capacity.

Under a benchmark policy, an incoming transfer request is rejected whenever $\eta X(t) + B(t) > \Gamma$, where η and Γ are tunable parameters. This class captures a wide range of commonly used threshold rules based on ED occupancy and boarding levels. For any fixed η, varying Γ traces out the achievable trade-off between congestion (measured by the average waiting time) and access (measured by transfer rejection rate) within this class of state-based policies. The resulting trade-off curve summarizes the attainable performance of policies that rely only on contemporaneous system information and do not use predictive inputs, providing a natural and fair benchmark for assessing the incremental value of prediction.

We evaluate benchmark policies for $\eta \in \{0, 0.5, 1\}$ and $\Gamma \in \{10, 11, \ldots, 100\}$. Once Γ is appropriately tuned, benchmark performance is largely insensitive to the choice of η (see Appendix D for details). For clarity, we therefore present results for $\eta = 0$.

For the proposed look-ahead policy, the numerical computation details for H_t are provided in Appendix D, and we construct an analogous trade-off curve by varying the ratio l/h, which governs the relative penalty placed on transfer rejection versus patient waiting.

Experimental design and baseline parameters. Unless otherwise noted, we consider a system with constant service capacity $c(t) = 30$ over a horizon $T = 150$. A fraction $\gamma = 0.2$ of arrivals correspond to transfer requests, with the remaining arrivals representing non-transfer patients. Service times

are exponential, with the service rate normalized to $\mu = 1$, so that time is measured in units of mean service time. The abandonment rate is $\theta = 0.1$, and patients are discharged directly from the ED with probability $p = 0.5$. The system is initialized at $(X(0), B(0)) = (30, 0)$. We normalize the waiting cost to $h = 1$ and vary the blocking cost $l \in [0, 10)$ to trace out the tradeoff curves. Admission decisions are updated periodically, with review intervals $\delta \in \{1, 5\}$ and $a = 0.5$ to compensate for the time lag between reviews. All reported performance measures are estimated as averages over 200 independent sample paths.

In the main paper, we fix the inpatient processing rate at $\nu(t) = 0.5$ to isolate the effect of predictive demand information. Appendix D considers scenarios with time-varying and reduced processing rates; across these settings, the qualitative comparison between policies remains unchanged. We assume the processing rate is known without error, i.e., $\widehat{\nu}(s \mid t) = \nu(s)$.

Arrival pattern and prediction error. We assess policy performance along two dimensions. First, we consider arrival patterns corresponding to a single congestion surge and two congestion surges, which capture distinct temporal structures of demand shocks. Second, and more importantly, we examine how different forms of prediction error in arrival rate forecasts affect policy performance.

In all congestion-surge scenarios, admission decisions are based on predicted arrival rates rather than perfect future information. Predicted arrival rates are generated on a discrete time grid with step size $\Delta t = 0.1$ and held constant between updates, with nonnegativity enforced by truncation.

5.1. Congestion surge scenarios

To isolate the value of forecast information from the details of any particular prediction method, we begin with a stylized baseline forecast-error model. At each review epoch, the policy is supplied with a forecast trajectory of future arrival rates obtained by perturbing the underlying rate trajectory with some unbiased noise. This provides a controlled setting for evaluating the policy under imperfect forecasts before considering richer forms of forecast misspecification in Section 5.2.

We examine two demand environments that differ in the temporal structure of congestion: a single-surge scenario and a two-surge scenario. In both cases, the baseline forecast errors are assumed to be unbiased and independently distributed. For each review epoch t, let $s_k = t + k\Delta t$, $k = 0, 1, \ldots$, denote the future forecast grid,

$$\widehat{\lambda}(s_k \mid t) = \lambda(s_k) + \epsilon_{t,k}, \quad \epsilon_{t,k} \sim \mathcal{N}(0, \sigma^2).$$

Thus, the forecasts preserve the underlying temporal pattern of demand while introducing noise in the predicted arrival rates.

5.1.1. Single congestion shock We first consider a scenario with a single congestion shock, where the arrival rate $\lambda(t)$ evolves as

$$\lambda(t) = \begin{cases} 2, & 0 \le t \le 50, \\ 20, & 50 < t \le 100, \\ 2, & 100 < t \le T. \end{cases}$$

In the simulation, we extend $\lambda(t) = 2$ beyond T. The left panel of Figure 2 shows a realization of the predicted arrival rate $\widehat{\lambda}(t \mid 0)$.

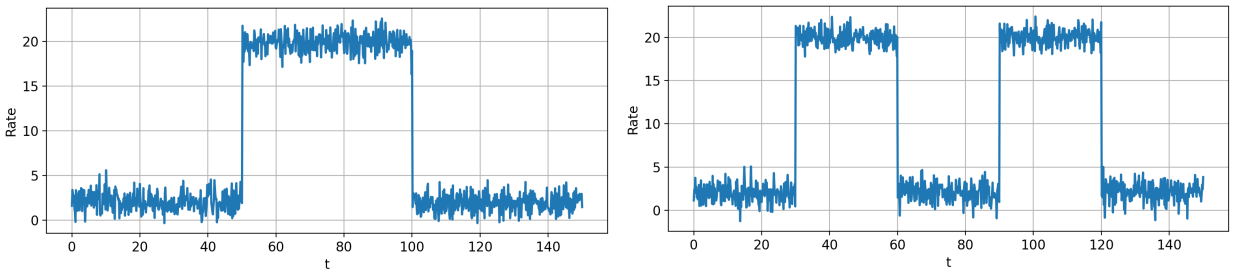


Figure 2 Left: single congestion shock. Right: two congestion shocks. Predicted arrival rate $\widehat{\lambda}(t \mid 0) = \lambda(t) + \epsilon_{0,t}$, where $\epsilon_{0,t} \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 1$.

Figure 3 compares the performance of the benchmark and look-ahead policies by plotting the trade-off between total patient waiting time and the proportion of rejected transfer requests. Across both panels, corresponding to review intervals $\delta = 1$ and $\delta = 5$ respectively, the look-ahead policy uniformly dominates the benchmark. Specifically, for any given level of waiting time, the look-ahead policy achieves a lower rejection rate, and conversely, for any given rejection rate, it yields shorter waiting times. These results highlight the operational value of incorporating predictive demand information into dynamic admission decisions, particularly in environments subject to transient congestion shocks.

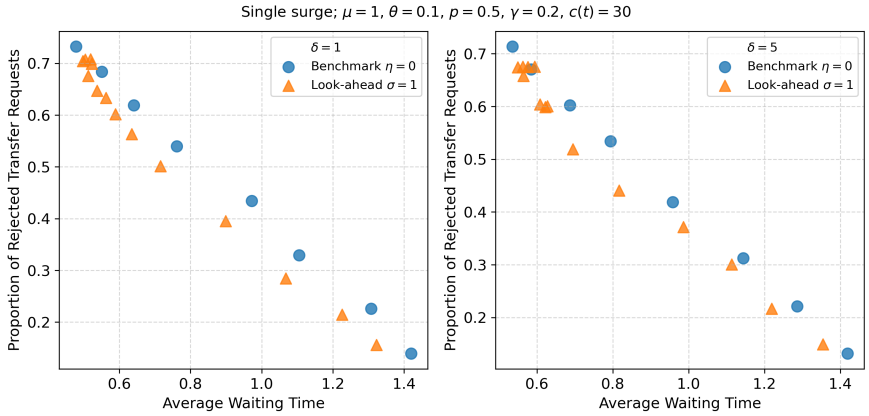


Figure 3 Single congestion shock: comparison between look-ahead v.s. benchmark.

5.1.2. Two congestion shocks We next consider two congestion shocks, with $\lambda(t) = 20$ for $t \in (30, 60] \cup (90, 120]$, $\lambda(t) = 2$ otherwise. The right panel of Figure 2 illustrates a realization of the predicted arrival rate $\widehat{\lambda}(t \mid 0)$.

Figure 4 compares the performance of the benchmark and look-ahead policies by plotting the trade-off between total patient waiting time and the fraction of rejected transfer requests. For both review intervals $\delta = 1$ and $\delta = 5$, the look-ahead policy again consistently dominates the benchmark. This demonstrates that the benefits of predictive admission control persist even in more complex environments with multiple congestion surges and intervening recovery periods.

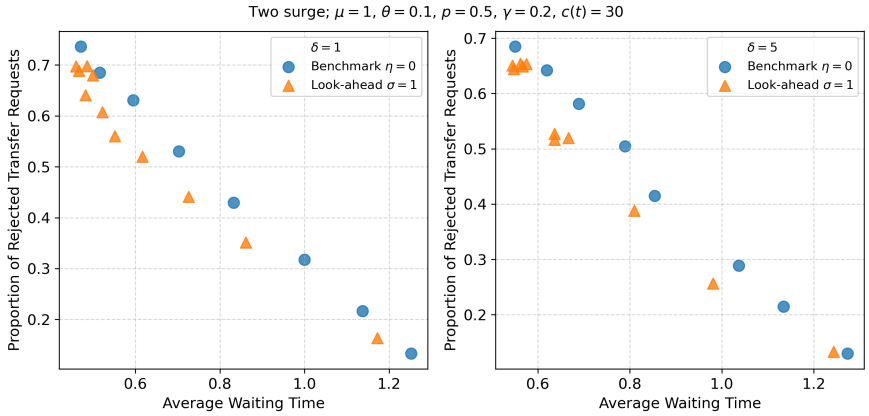


Figure 4 Two congestion shocks: comparison between look-ahead v.s. benchmark.

5.2. Prediction Errors

In Section 5.1, we evaluated the policy under a baseline forecast-error model with unbiased, independently distributed noise in the predicted arrival rates. We now examine robustness to a broader range of forecast misspecification.

Rather than deriving forecast errors from a particular prediction algorithm, we specify the error process $e_\lambda(s \mid t) = \widehat{\lambda}(s \mid t) - \lambda(s)$ directly. This allows us to vary distinct dimensions of forecast quality one at a time, including noise magnitude, temporal persistence, horizon dependence, systematic bias, and timing error, and isolate their effects on admission-control performance. In this way, we can assess the operational consequences of different forms of forecast error without confounding them with the choice of a particular prediction method. Accordingly, the experiments in this section should be interpreted as controlled forecast-error stress tests rather than as an evaluation of any specific forecasting algorithm. Section 6 complements this analysis with an end-to-end implementation in which forecasts are generated by models trained and evaluated on hospital data.

Recall that for each review epoch t, $s_k = t + k\Delta t$, $k = 0, 1, \ldots$, is the future forecast grid. We specify the forecast-error process $e_\lambda(s_k \mid t)$ directly.

Scenario I: *Independent forecast errors (baseline).*

$e_\lambda(s_k \mid t) \sim \mathcal{N}(0, \sigma^2)$. We consider $\sigma \in \{0, 1, 2, 4\}$ to assess sensitivity to the noise level.

Scenario II: *Correlated forecast errors.*

$e_\lambda(s_k \mid t) = \rho e_\lambda(s_{k-1} \mid t) + u_{t,k}$, $u_{t,k} \sim \mathcal{N}(0, \sigma_u^2)$, where $\rho \in \{0.7, 0.99\}$ and $\sigma_u^2 = (1 - \rho^2)\sigma^2$.

Scenario III: *Horizon-dependent forecast uncertainty.*

$e_\lambda(s_k \mid t) \sim \mathcal{N}(0, \sigma^2(1 + \beta k))$. We consider $\beta \in \{0.02, 0.2\}$.

Scenario IV: *Biased forecasts.*

$e_\lambda(s_k \mid t) = \alpha\lambda(s_k) + \epsilon_{t,k}$, $\epsilon_{t,k} \sim \mathcal{N}(0, \sigma^2)$, i.e., $\widehat{\lambda}(s_k \mid t) = (1 + \alpha)\lambda(s_k) + \epsilon_{t,k}$. We consider $\alpha \in \{-0.6, -0.4, -0.2, 0.2, 0.4, 0.6\}$. Positive values of α correspond to persistent overestimation of demand, while negative values represent underestimation.

Scenario V: *Shock timing misprediction.*

$e_\lambda(s_k \mid t) = \lambda(s_k - \xi) - \lambda(s_k) + \epsilon_{t,k} \quad \epsilon_{t,k} \sim \mathcal{N}(0, \sigma^2)$, i.e., $\widehat{\lambda}(s_k|t) = \lambda(s_k - \xi) + \epsilon_{t,k}$. We consider $\xi \in \{5, 10\}$.

For Scenarios II–V, we consider $\sigma \in \{1, 2\}$. For ease of exposition, the main paper reports results for the two-surge scenario with $\sigma = 2$ and $\delta = 1$. Results for additional cases, including the single-surge scenario and alternative σ and δ, are qualitatively similar and reported in Appendix D.

Across the baseline specification (Scenario I) and Scenarios II–IV with positive α, the performance of the look-ahead policy remains qualitatively stable. As shown in Figure 5, we observe little systematic deterioration in the trade-off between patient waiting and transfer rejection under unbiased noise, persistent (correlated) forecast errors, horizon-dependent uncertainty, or systematic overprediction of demand. These results suggest that the policy is relatively insensitive to forecast errors that perturb the predicted rate trajectory without substantially obscuring the magnitude or timing of impending congestion.

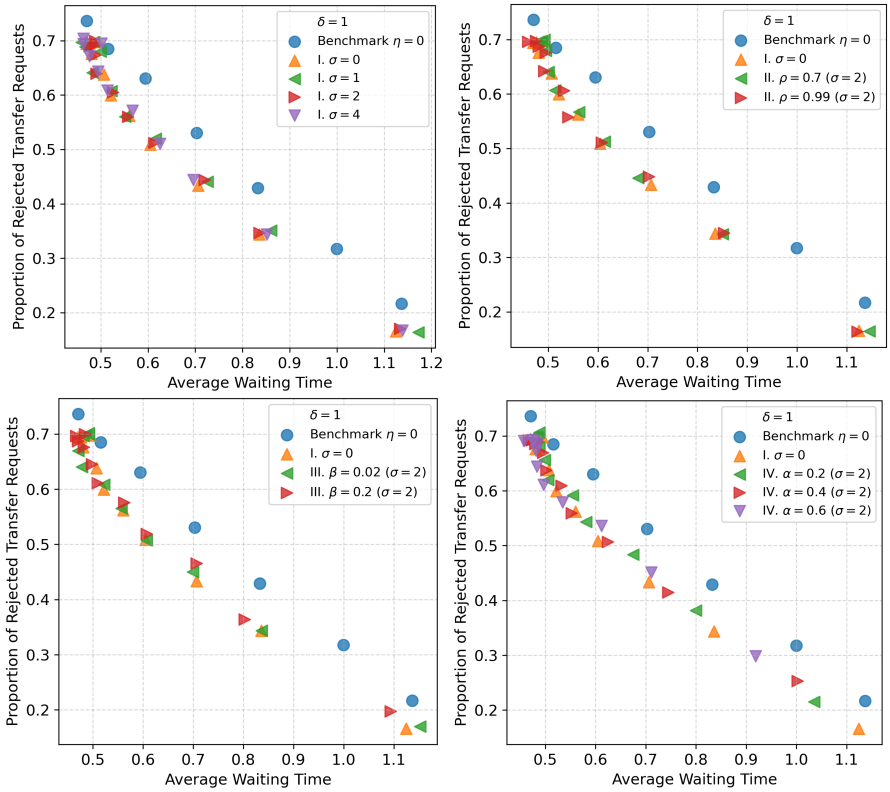


Figure 5 Similar performance with prediction errors in scenarios I, II, III, IV with positive α

In contrast, performance deteriorates under systematic underprediction of demand (Scenario IV with negative α) and errors in surge timing (Scenario V), as illustrated in Figure 6. Underprediction causes the policy to underestimate the severity of future congestion, while delayed surge predictions prevent it from responding sufficiently early. These results highlight an important asymmetry in the operational consequences of forecast errors: errors that obscure the magnitude or timing of

impending congestion are substantially more consequential than moderate noise that preserves these features of the future demand trajectory.

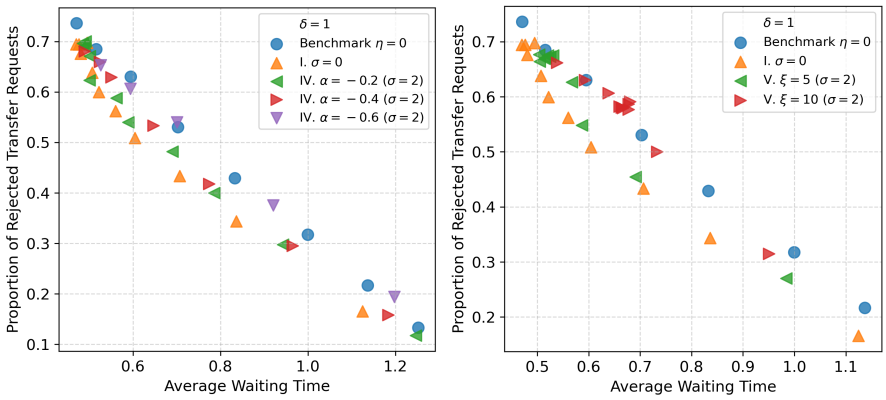


Figure 6 Degradation in performance with prediction errors in scenarios IV (with negative α) and V

6. Case Study: Implementation with Real Hospital Data

In this section, we evaluate the proposed look-ahead admission control policy using operational data from a large academic medical center. The purpose of this case study is to assess whether the policy can be implemented with realistically available information, including data-driven forecasts of demand and boarding delays, and whether it yields measurable improvements relative to the hospital's current transfer-admission practice when calibrated to empirical operating conditions.

A central practical consideration is that transfer patients differ systematically from non-transfer patients in their likelihood of requiring inpatient admission. To capture this empirically important heterogeneity while preserving the policy logic developed earlier, Section 6.1 generalizes the base model to allow type-dependent admission probabilities. This extension modifies only the fluid dynamics used to compute the hitting time; the admission decision remains a threshold rule in the predicted hitting time.

Building on this implementation model, Section 6.2 describes the data sources, forecasting models for arrivals and processing rates, and parameter calibration. Section 6.3 compares the performance of the look-ahead policy to the hospital's current policy using simulated counterfactuals over the study period. Together, the results illustrate how predictive information can be operationalized through a periodic-review rule by quantifying the improvements in the congestion-access trade-off.

6.1. Operational Model with Heterogeneous Admission Outcomes

To support implementation, we generalize the base model by allowing the probability of inpatient admission to depend on patient type. Specifically, patients are classified into external patients (type 1) and transfer patients (type 2). Let p_1 and p_2 denote the probabilities that type-1 and type-2 patients, respectively, are discharged directly after completing ED service. This distinction matters because transfer patients are typically more likely to require inpatient admission.

The stochastic primitives are as in Section 2; in particular, arrivals, service times, abandonment behavior, and boarding dynamics are unchanged. We additionally assume that transfer patients are prioritized for ED service whenever capacity is available, which reflects common operational practice when transfer patients arrive through coordinated pathways.

For implementation, we work with the corresponding fluid approximation. Let $x_i(t)$, $z_i(t)$, and $q_i(t)$ denote the fluid analogues of the number of type-i patients in system, in service, and waiting in queue, respectively, where $q_i(t) = x_i(t) - z_i(t)$, for $i \in \{1, 2\}$. Let $b(t)$ denote the boarding fluid. The fluid dynamics are

$$\begin{aligned} \dot{x}_1(t) &= \lambda_1(t) - \mu z_1(t) - \theta q_1(t),\ \dot{x}_2(t) = g(t)\lambda_2(t) - \mu z_2(t) - \theta q_2(t), \\ \dot{b}(t) &= (1-p_1)\mu z_1(t) + (1-p_2)\mu z_2(t) - \nu(t) b(t), \end{aligned} \tag{11}$$

where $g(t) \in [0, 1]$ denotes the (fluid) transfer acceptance control. Service capacity is allocated according to transfer-priority:

$$z_2(t) = \min\{c(t) - b(t),\, x_2(t)\},\ z_1(t) = \min\{c(t) - b(t) - z_2(t),\, x_1(t)\}. \tag{12}$$

We retain the hitting-time structure of the policy in Section 3 as a heuristic for this heterogeneous model. Let $H_t(x_1, x_2, b)$ denote the time until both fluid queues clear, starting from (x_1, x_2, b) at time t under full acceptance $g(s) \equiv 1$ for $s \geq t$. We implement a periodic-review policy: at review time $k\delta$, we compute $\widehat{H}_{k\delta}$ using the current state $(X_1(k\delta), X_2(k\delta), B(k\delta))$ and the forecasted rates, and hold the admission decision fixed throughout $[k\delta, (k+1)\delta)$. Specifically, transfer requests are blocked during this interval if

$$\widehat{H}_{k\delta}\big(X_1(k\delta), X_2(k\delta), B(k\delta)\big) - \frac{\delta}{2} > -\frac{1}{\theta}\log\left(1 - \frac{l\theta}{h}\right), \tag{13}$$

and accepted otherwise. The shift term $\delta/2$ is a simple correction for within-window state evolution under periodic review.

6.2. Data, Prediction, and Model Calibration

We conduct the case study using detailed ED patient flow data, including some inter-facility transfer requests, from April 2022 to November 2024 at a large quaternary-care academic ED in the northeastern United States. The data contain time-stamped records of ED arrival, physician first contact, inpatient bed request, and ED departure, which allow us to reconstruct time-varying arrival patterns, service dynamics, and boarding delays. To support forecasting and calibration, we additionally collect and construct time-varying measures of inpatient occupancy (including non-ED admissions), extract the hospital's recorded transfer-admission status, and merge local weather variables (temperature, precipitation, snowfall, and wind speed). Additional details on data processing are provided in Appendix C.1.

The resulting dataset contains 311,739 ED encounters (301,586 external arrivals, 6,826 completed ED-to-ED transfers, and 3,327 rejected ED-to-ED transfers), of which 21.04% involve a request

for inpatient admission. We define boarding time as the duration between inpatient bed request and ED departure. On average, the ED experiences 13.32 arrivals per hour. Boarding times have a mean of 17.88 hours, and their empirical distribution (Figure 7) exhibits substantial dispersion and a pronounced right tail.

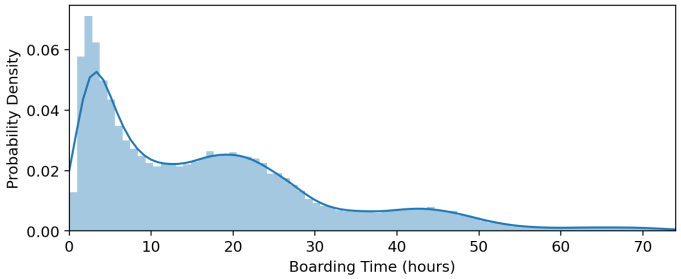


Figure 7 Histogram of boarding time in hours

We first instantiate the forecasting framework of Section 4.2 using statistical and machine-learning models trained on historical hospital data. The predictive accuracy reported below is based on one-step-ahead forecasts using observed lagged responses, whereas policy implementation uses recursively generated multi-step forecasts, as described below.

Arrival-rate prediction. We forecast arrivals at a two-hour resolution using information available at the beginning of each interval (see Appendix C.3 for details of arrival patterns). The predictors include: (i) calendar variables (month, day-of-week, time-of-day, holiday indicators); (ii) lagged arrival counts over the preceding one to seven intervals; and (iii) weather variables (temperature, precipitation, snowfall, wind, and indicators for extreme heat). For weather inputs, we use realized weather as a proxy for short-term forecasts available at the review epoch, motivated by the availability of such forecasts in practice (e.g., Bauer et al. 2015, Lam et al. 2023, and Dowell et al. 2022). This approximation neglects weather forecast errors over the prediction horizon. Furthermore, we split the data chronologically into equally sized training and test samples and fit both linear regression and boosted tree models (estimated with XGBoost). Table 1 reports their predictive accuracy; both models achieve similar test-set RMSE (5.69 for linear regression and 5.66 for XGBoost), suggesting similar out-of-sample prediction accuracy.

Dataset	**Linear Regression**	**XGBoost**
Train	5.62	4.29
Test	5.69	5.66

Table 1 RMSE values for arrival count prediction models.

Processing-rate estimation and prediction. We next estimate and predict the processing rate governing transfers from the ED to inpatient units. We assume boarding times are exponential and the processing rate $\nu(t)$ is piecewise constant over prespecified 8-hour intervals, partitioning each 24-hour day into three 8-hour periods. For a given interval $[t_k, t_{k+1})$, we include all patients who

spend any time boarding during that interval. Each patient contributes exposure from the later of their boarding start time and t_k until either (i) the patient completes boarding and leaves the ED for an inpatient unit within the interval, or (ii) the interval ends, in which case the observation is right-censored at t_{k+1}. Thus, patients who complete boarding contribute both a transfer-completion event and their time at risk, whereas patients who remain boarding at the end of the interval contribute only their accumulated time at risk. Under the exponential assumption, the likelihood therefore depends on the number of completed transfers and the total boarding exposure accumulated during the interval. Maximizing this likelihood yields the closed-form estimator $\nu_k = (\#\text{completions})/(\text{total exposure})$. We then set $\nu(t) = \nu_k$ for $t \in [t_k, t_{k+1})$ (see Appendix C.2 for details). The average estimated processing rate is 0.06.

We then use the interval-specific estimates ν_k as the response variable for forecasting future inpatient processing rates. Predictors include temporal indicators, lagged estimate of the processing rate, and weather variables. As with the arrival rate forecast, we split the data chronologically into equally sized training and test periods to preserve the temporal ordering of the observations, and fit both linear regression and XGBoost models on the training sample. Table 2 reports resulting predictive accuracy. The two models achieve similar out-of-sample performance, with test-set RMSEs of approximately 0.02.

Dataset	**Linear Regression**	**XGBoost**
Train	0.02	0.01
Test	0.02	0.02

Table 2 RMSE values for processing rate prediction models.

Model calibration and simulation. We calibrate a multi-server queueing simulator to reproduce key features of observed ED operations over the study period. Arrivals follow a time-varying Poisson process with piecewise-constant rates over two-hour intervals, estimated directly from the data. The observed fraction of arrivals classified as transfer patients is $\gamma = 0.033$. Discussions with our clinical partners suggest, however, that transfer requests may be substantially underrecorded in the administrative data; for example, declined requests may not be systematically entered into transfer logs. We therefore treat $\gamma = 0.033$ as a conservative baseline scenario and consider larger values to assess settings with greater transfer exposure, consistent with rates observed elsewhere (Moore et al. 2023). Patient patience times are assumed to be exponentially distributed, with abandonment rate estimated by maximum likelihood as $\theta = 0.0353$. Service times are also assumed exponential, with an estimated mean of 8.1408 hours, corresponding to a service rate $\mu = 1/8.1408$. We set the number of servers to $c(t) = 154$, equal to the median number of patients simultaneously in service during the study period. The probability of direct discharge after ED service is estimated separately for external and transfer patients, yielding $p_1 = 0.7927$ and $p_2 = 0.5494$, respectively. Patients who are not discharged directly enter the boarding stage, with transfer to inpatient units

governed by the estimated time-varying processing rate $\nu(t)$. To improve numerical stability while preserving the empirical tail behavior, we truncate service and boarding times at their respective observed maxima of 696 and 155 hours.

In implementation, at each review epoch t, we generate forecast trajectories $\widehat{\lambda}(s \mid t)$ and $\widehat{\nu}(s \mid t)$, $s \geq t$, and use them to evaluate the hitting-time functional. Because both models include lagged response variables, we generate multi-step forecasts recursively, substituting earlier predictions whenever the required lagged responses are not yet observed. Apart from the realized-weather approximation described above, these forecasts use only information available at the review epoch. For the two arrival streams, we set $\widehat{\lambda}_1(s \mid t) = (1-\gamma)\widehat{\lambda}(s \mid t)$, $\widehat{\lambda}_2(s \mid t) = \gamma\widehat{\lambda}(s \mid t)$. We vary the cost ratio l/h over its feasible range $0 < l/h < 1/\theta$ and consider review intervals δ ranging from one hour to twelve hours. The parameter grid considered is

$$\frac{l}{h} \in \{0.5, 1, 4, 6, 8, 10, 12, 16, 20, 24, 26\},\ \delta \in \{1, 2, 4, 8, 12\} \text{ hours},\ \gamma \in \{0.033, 0.1, 0.2\}. \tag{14}$$

For each configuration, we report the fraction of rejected transfer requests, average waiting time, and average boarding census. To mitigate transient effects, we compute performance after excluding the first and last months of each simulated period.

6.3. Performance Evaluation

The hospital's current policy blocks all transfer requests whenever the boarding census exceeds a fixed threshold $\bar{b}$. During the study period, this threshold was approximately 45, and the observed boarding census exceeded it 60.9% of the time. To benchmark this policy, we simulate the system under thresholds $\bar{b} \in \{41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65\}$. With $\bar{b} = 45$ an hourly review interval ($\delta = 1$), transfers are blocked for 54.8%, 58.3%, and 59.3% of the simulated time when $\gamma = 3.3\%$, 10%, and 20%, respectively. These proportions are reasonably close to the observed value. Note that the proportion of time transfers are blocked differs from the fraction of transfer requests rejected, reported below, because transfer arrival rates vary over time.

We evaluate the look-ahead policy (13) over the parameter grid in (14). Under the hospital's boarding-threshold policy, increasing $\bar{b}$ reduces transfer rejection at the expense of longer patient waiting times. Increasing the cost ratio l/h under the look-ahead policy produces a similar trade-off. This congestion–access trade-off is less pronounced when $\gamma = 3.3\%$, because transfers account for a small share of arrivals and rejecting them has limited impact on aggregate congestion. Detailed results for each policy, including boarding census and sensitivity to the review interval, are reported in Appendix C.4.

Figure 8 compares the congestion–access trade-offs achieved by the two policies, using a one-hour review interval for the look-ahead policy. For both $\gamma = 0.10$ and $\gamma = 0.20$, the look-ahead policy achieves a more favorable congestion–access trade-off, yielding lower transfer rejection rates

for comparable or shorter patient waiting times. For example, when $\gamma = 0.20$ and the average waiting time is capped at 1 hours, the look-ahead policy (with $l/h = 8$) reduces the rejection fraction from 0.447 (under the policy $\bar{b} = 51$) to 0.391, corresponding to a 12% reduction in declines (equivalently, 10% increase in acceptances). Conversely, when the rejection fraction is constrained to be no more than 0.30, the look-ahead policy (with $l/h = 16$) reduces average waiting time from 1.702 hours under the hospital policy ($\bar{b} = 59$) to 1.348 hours, a reduction of approximately 21%. These comparisons demonstrate that incorporating forecasts of future operating conditions into a simple periodic-review rule can meaningfully improve the congestion–access trade-off relative to current practice.

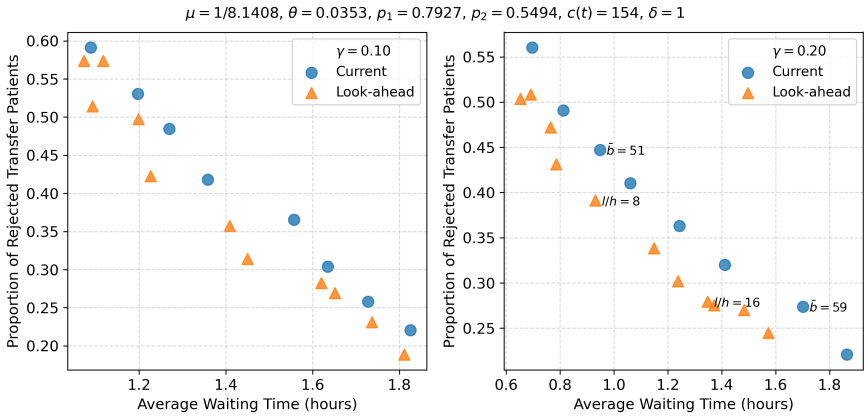


Figure 8 Comparison between Look-ahead Policy versus Current Policy

Additional forecasting specification. Appendix C.5 examines an expanded forecasting specification that adds ED census to the arrival-rate model and boarding census and service-level inpatient occupancy to the processing-rate model. In this exercise, we use the realized future values of these additional predictors over the look-ahead horizon. Because these values are unavailable at the review epoch, an operational implementation would require additional models to forecast them. This exercise therefore serves as a sensitivity analysis with additional information, rather than an implementable forecasting specification. Predictive accuracy and policy performance are similar to those under the main specification.

7. Conclusion

We develop a proactive admission control framework for emergency care settings that integrates predictive information on patient arrivals and inpatient admission delays. As crowding increasingly threatens care quality and contributes to staff burnout, our approach provides a practical, data-driven mechanism for improving patient flow and system performance.

To address the operational complexity of admission decisions under transient congestion, we adopt a fluid approximation and characterize the optimal fluid control. Guided by this analysis,

we propose a look-ahead admission policy that is intuitive, easy to implement, and asymptotically optimal as demand and capacity scale.

Extensive synthetic experiments and a data-calibrated case study based on real hospital data demonstrate the effectiveness of the proposed approach. Across a wide range of operating conditions, the look-ahead policy consistently outperforms non-predictive benchmarks by achieving better trade-offs between patient waiting times and transfer rejections, enabling hospitals to improve patient access (i.e., decrease transfer rejections) while maintaining waiting times.

The magnitude of these benefits depends on both transfer-request volume and congestion. The policy is particularly relevant to tertiary and referral hospitals that receive substantial transfer demand and experience severe ED crowding. When transfer requests account for a small share of demand, the potential reduction in aggregate ED congestion is more limited, although managing these requests may still be useful during periods of acute crowding.

Several directions merit further investigation. First, the framework could be extended to jointly manage inbound and outbound transfers. Although outbound ED-to-ED transfers are rare at our study hospital, which offers a broad range of clinical services, they may provide an additional means of managing congestion at other hospitals. Direct ED-to-inpatient transfers offer another option: patients who have completed their ED evaluation and require admission may be transferred directly to an available inpatient bed at another hospital. When treated as exogenous, these departures can be incorporated into the boarding processing rate $\nu(t)$. Explicitly controlling them would require accounting for bed availability at receiving hospitals, transfer costs, and clinical eligibility. A network extension could examine how hospitals coordinate these decisions to improve access and reduce congestion across the system.

Second, a prospective pilot would help assess the operational value of the proposed policy. Our approach is designed to facilitate implementation: it combines forecasts of arrival and inpatient processing rates with an interpretable decision rule applied at regular review intervals. Its cost ratio provides a transparent way to adjust the balance between congestion and transfer access. A good evaluation would help identify appropriate review intervals and procedures for updating forecasts as operating conditions change.

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Online Appendix

Appendix A: Proof of the Fluid Optimal Control (Theorems 1–2)

In this section, we prove the optimal fluid admission control policy via Pontryagin's Minimum Principle. We first state a version adapted to our problem and then verify its sufficient conditions.

Fix any admissible $g(\cdot)$. It is without loss of optimality to enforce work conservation, which implies the service rate $z(t)$ must satisfy: $z(t) = \min\{x(t), c(t) - b(t)\}$. Equivalently, the service allocation satisfies $z(t) \le x(t)$ and $z(t) \le c(t) - b(t)$, with at least one of these inequalities binding at each time t. For the purpose of deriving the optimal control, we define the **Hamiltonian** $\mathcal{H}$ representing the system dynamics and costs, and the **Augmented Lagrangian** $\mathcal{L}$ which incorporates the boundary constraints via multipliers.

Define the Hamiltonian $\mathcal{H} : [0,\tau] \times \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}$ as: $\mathcal{H}(t,x,b,g,p_x,p_b) := p_x\Big[\lambda_1(t) + g\lambda_2(t) - \mu z - \theta(x-z)\Big] + p_b\Big[(1-p)\mu z - \nu(t)b\Big] + h(x-z) + l\lambda_2(t)(1-g)$. To handle the nonsmooth dynamics $z = \min\{x, c-b\}$, we introduce the Lagrangian multipliers $\eta_1(t), \eta_2(t) \ge 0$ corresponding to the capacity and queue constraints, respectively. The Augmented Lagrangian is: $\mathcal{L}(t,x,b,g,z,p,\eta) := \mathcal{H} - \eta_1\big(c(t) - b - z\big) - \eta_2(x-z)$.

The following sufficiency theorem is adapted from standard results on optimal control with state constraints (e.g., (Seierstad and Sydsæter 1987, Hartl et al. 1995)).

THEOREM 3. *Consider the optimal control problem* (5)*. Let* (x^*, b^*, g^*, z^*) *be a feasible work-conserving trajectory with final clearing time* τ^**. Suppose there exist absolutely continuous adjoint functions* $p_x(\cdot)$*,* $p_b(\cdot)$ *and piecewise continuous multipliers* $\eta_1(\cdot), \eta_2(\cdot) \ge 0$ *on* $[0,\tau^*]$ *such that for almost all* t*:*

1. ***Convexity:*** *The map* $(x,b,g) \mapsto \mathcal{H}\big(t,x,b,g,p_x,p_b\big)$ *is convex on* $\mathbb{R}^2 \times [0,1]$*.*
2. ***Adjoint Equations:*** *The time derivatives of the adjoint variables satisfy:* $-\dot{p}_x(t) = \frac{\partial \mathcal{L}}{\partial x}$*,* $-\dot{p}_b(t) = \frac{\partial \mathcal{L}}{\partial b}$*, evaluated along the candidate trajectory.*
3. ***Stationarity & Complementarity:*** *The multipliers satisfy* $\frac{\partial \mathcal{L}}{\partial z} = 0, \eta_1(c - b^* - z^*) = 0, \eta_2(x^* - z^*) = 0$*.*
4. ***Minimization Condition:*** $g^*(t) \in \arg\min_{0 \le g \le 1} \mathcal{H}\big(t, x^*(t), b^*(t), g, p_x(t), p_b(t)\big)$*.*
5. ***Transversality:*** $p_x(\tau^*) = 0,\ p_b(\tau^*) = 0$*.*

Then $(x^*(\cdot), b^*(\cdot), g^*(\cdot))$ *is an optimal solution.*

REMARK 1. We compute the adjoint variables using phasewise Hamiltonians. On a congested phase, $z = c - b$; on a strictly uncongested phase, $z = x$. Denote the corresponding affine Hamiltonians by $\mathcal{H}_c$ and $\mathcal{H}_u$. The global sufficiency conditions are verified as follows. Define $A(t) := h + (\mu - \theta)p_x(t) - (1-p)\mu p_b(t)$. Since $\mathcal{H}_c - \mathcal{H}_u = A(t)(x + b - c(t))$, the work-conserving Hamiltonian satisfies $\mathcal{H} = \mathcal{H}_u + A(t)(x + b - c(t))^+$. Thus $A(t) \ge 0$ implies global convexity. Moreover, setting

$$(\eta_1(t), \eta_2(t)) = \begin{cases} (A(t), 0), & x^*(t) + b^*(t) > c(t), \\ (0, A(t)), & x^*(t) + b^*(t) < c(t) \end{cases}$$

gives nonnegative multipliers satisfying stationarity and complementarity, because $\partial\mathcal{L}/\partial z = -A + \eta_1 + \eta_2 = 0$. At an isolated boundary contact, any nonnegative split with $\eta_1 + \eta_2 = A$ may be used. With these multipliers, the augmented adjoint equations reduce exactly to the phasewise adjoint equations. Accordingly, the phasewise calculations below, together with the verification of $A(t) \ge 0$,

establish the sufficient conditions globally, without requiring competing trajectories to share the candidate's congestion periods.

A.1. Proof of Theorem 1

We derive the candidate from the congested-phase optimality conditions and then verify Theorem 3.

Under Assumption 1, the fluid queue remains positive before the clearing time τ and empties at τ; in particular, $q(t) > 0$ for $t \in [0, \tau)$ and $q(\tau) = 0$. Hence $z(t) = c(t) - b(t)$ on this interval. The congested-phase Hamiltonian (with $z = c - b$ already substituted) is $\mathcal{H}_c(t, x, b, g, p_x, p_b) := p_x\Big[\lambda_1(t) + g\lambda_2(t) - \mu(c(t) - b) - \theta(x + b - c(t))\Big] + p_b\Big[(1-p)\mu(c(t) - b) - \nu(t)b\Big] + h(x + b - c(t)) + l\lambda_2(t)(1 - g)$. The adjoint equations are $-\dot{p}_x(t) = \frac{\partial \mathcal{H}_c}{\partial x} = -\theta p_x(t) + h$, $-\dot{p}_b(t) = \frac{\partial \mathcal{H}_c}{\partial b} = (\mu - \theta)p_x(t) - \big((1-p)\mu + \nu(t)\big)p_b(t) + h$. We construct the adjoint backward from $p_x(\tau) = p_b(\tau) = 0$, as required by Theorem 3 with $\tau^* = \tau$. In particular, solving the first adjoint equation gives

$$p_x(t) = \frac{h}{\theta}\Big(1 - e^{-\theta(\tau - t)}\Big), \qquad t \in [0, \tau]. \tag{15}$$

(When $\theta = 0$, the same calculation yields $p_x(t) = h(\tau - t)$.)

Next, Since $\partial \mathcal{H}_c / \partial g = \lambda_2(t)(p_x(t) - l)$, pointwise minimization gives acceptance when $p_x(t) < l$ and rejection when $p_x(t) > l$, wherever $\lambda_2(t) > 0$. At equality, any control in $[0, 1]$ minimizes the Hamiltonian. We choose the threshold convention

$$g^*(t) = \mathbf{1}\{p_x(t) \le l\} \quad \text{for a.e. } t \in [0, \tau]. \tag{16}$$

Recall that $H_t(x, b)$ is the time to clear under full acceptance, starting from (x, b) at time t. Let $f(u) = h(1 - e^{-\theta u})/\theta$ and define $K = \inf\{u \ge 0 : f(u) \ge l\}$, with $\inf \varnothing = \infty$. If $l \ge h/\theta$, then $K = \infty$ and (16) prescribes full acceptance, as does the hitting-time rule. Otherwise, $K = f^{-1}(l)$. By (15), $p_x(t)$ is strictly decreasing. Writing $\zeta = \max\{0, \tau - K\}$, the candidate therefore rejects for $t < \zeta$ and accepts for $t \ge \zeta$, with acceptance from the outset when $\zeta = 0$.

Phase A: $t \in [\zeta, \tau]$. The candidate accepts all transfers from t onward. Thus, its trajectory coincides with the all-accept trajectory used to define H_t, giving $H_t(x(t), b(t)) = \tau - t$. Together with (15), this yields $p_x(t) = f(H_t(x(t), b(t)))$ on this phase. Hence the hitting-time rule agrees with (16).

Phase B: $t \in [0, \zeta)$. The candidate rejects transfers until ζ, whereas H_t is computed under full acceptance. As long as both trajectories are congested, their boarding states coincide, and full acceptance produces a treatment population no smaller than that under the candidate. It therefore cannot clear earlier. Consequently, $H_t(x(t), b(t)) \ge \tau - t > \tau - \zeta = K$. Thus $f(H_t(x(t), b(t))) > l$, and the hitting-time rule also rejects on this phase.

It remains to verify $A(t) \ge 0$ in Remark 1. Let $R(t) = p_x(t) - (1-p)p_b(t)$ and $\beta(t) = (1-p)\mu + \nu(t)$. The congested-phase adjoint equations imply

$$-\dot{R}(t) + \beta(t)R(t) = p\big(h - \theta p_x(t)\big) + \nu(t)p_x(t), \quad R(\tau) = 0.$$

By (15), the right-hand side is nonnegative, so backward integration gives $R(t) \ge 0$ on $[0, \tau]$. Therefore, $A(t) = h - \theta p_x(t) + \mu R(t) > 0$. The multipliers in Remark 1 thus satisfy stationarity and complementarity, and the global Hamiltonian is convex. The feasible candidate and its adjoint satisfy the adjoint equations and transversality, and g^* defined in (16) minimizes the Hamiltonian. Theorem 3 therefore establishes that the control g^* is optimal for (5). $\square$

A.2. Proof of Theorem 2

Recall that $I_1 = [0, \tau_1]$ and $I_2 = [\sigma_2, \tau_2]$ are the two congestion periods, separated by the uncongested gap $G_1 = [\tau_1, \sigma_2]$. If there is no gap, the two episodes form a single congested interval and the preceding single-episode argument applies. We therefore consider a nontrivial uncongested gap.

The proof constructs the adjoint variables by backward integration and verifies Theorem 3 in three phases: the first congested period $[0, \tau_1)$, the uncongested period (τ_1, σ_2), and the second congested period (σ_2, τ_2). We construct the adjoint backward from $p_x(\tau_2) = p_b(\tau_2) = 0$ and apply Theorem 3 with $\tau^* = \tau_2$.

Costate continuity at regime-switching times. We construct the adjoint as an absolutely continuous function, matching the phasewise solutions continuously at τ_1 and σ_2. The switching surface $x + b - c(t) = 0$ separates two service regimes; the fluid vector field and running cost agree across this surface. Since boundary contacts are isolated, the phasewise adjoint equations hold almost everywhere. The multiplier construction in Remark 1, verified below, then ensures that these equations satisfy the global verification conditions.

Phase 3 ($t \in [\sigma_2, \tau_2)$): Second Congested Period. Since $q^*(t) = x^*(t) + b^*(t) - c(t) > 0$, the service rate is $z^*(t) = c(t) - b^*(t)$. The congested-phase Hamiltonian is $\mathcal{H}_c(t, x, b, g, p_x, p_b) := p_x\Big[\lambda_1(t) + g\lambda_2(t) - \mu(c(t) - b) - \theta(x + b - c(t))\Big] + p_b\Big[(1-p)\mu(c(t) - b) - \nu(t)b\Big] + h(x + b - c(t)) + l\lambda_2(t)(1 - g)$. The adjoint equations are $-\dot{p}_x(t) = \frac{\partial \mathcal{H}_c}{\partial x} = -\theta p_x(t) + h$, $-\dot{p}_b(t) = \frac{\partial \mathcal{H}_c}{\partial b} = (\mu - \theta)p_x(t) - \big((1-p)\mu + \nu(t)\big)p_b(t) + h$. With $p_x(\tau_2) = 0$, we obtain $p_x(t) = \frac{h}{\theta}\left(1 - e^{-\theta(\tau - t)}\right)$. For p_b, let $\beta(t) = (1-p)\mu + \nu(t)$. With $p_x(\tau_2) = p_b(\tau_2) = 0$, we have

$$p_b(t) = \int_t^{\tau_2} [h + (\mu - \theta)p_x(s)] \exp\left(-\int_t^s \beta(u)du\right) ds. \tag{17}$$

Phase 2 ($t \in [\tau_1, \sigma_2)$): Uncongested Period. In this phase, $q^*(t) = 0$ and $z^*(t) = x^*(t)$. The uncongested-phase Hamiltonian is therefore $\mathcal{H}_u(t, x, b, g, p_x, p_b) := p_x\big[\lambda_1(t) + g\lambda_2(t) - \mu x\big] + p_b\big[(1-p)\mu x - \nu(t)b\big] + l\lambda_2(t)(1 - g)$. The adjoint equations are

$$-\dot{p}_x(t) = \frac{\partial \mathcal{H}_u}{\partial x} = -\mu p_x(t) + (1-p)\mu p_b(t), \tag{18}$$

$$-\dot{p}_b(t) = \frac{\partial \mathcal{H}_u}{\partial b} = -\nu(t)p_b(t). \tag{19}$$

The boundary values at $t = \sigma_2$ are inherited continuously from Phase 3. In particular, integrating (19) backward from σ_2, we have $p_b(t) = p_b(\sigma_2)\exp\big(-\int_t^{\sigma_2}\nu(u)\,du\big)$. Integrating (18) backward from σ_2, we have $p_x(t) = e^{-\mu(\sigma_2 - t)}p_x(\sigma_2) + (1-p)\mu\int_t^{\sigma_2} e^{-\mu(u-t)}\exp\big(-\int_u^{\sigma_2}\nu(r)\,dr\big)\,du\,p_b(\sigma_2)$.

Phase 1 ($t \in [0, \tau_1)$): First Congested Period. As in Phase 3, the system is congested and the Hamiltonian is $\mathcal{H}_c$. Thus the adjoint equations again satisfy $-\dot{p}_x(t) = -\theta p_x(t) + h$, $-\dot{p}_b(t) = (\mu - \theta)p_x(t) - \big((1-p)\mu + \nu(t)\big)p_b(t) + h$. The values at $t = \tau_1$ are obtained by matching to Phase 2. In particular, we have $p_x(t) = p_x(\tau_1)e^{-\theta(\tau_1 - t)} + \frac{h}{\theta}\left(1 - e^{-\theta(\tau_1 - t)}\right)$. and $p_b(t) = e^{-\int_t^{\tau_1}\beta(s)ds}p_b(\tau_1) + \int_t^{\tau_1} e^{-\int_t^s \beta(u)du}\,[h + (\mu - \theta)p_x(s)]\,ds$.

Since $\frac{\partial \mathcal{H}_c}{\partial g} = \frac{\partial \mathcal{H}_u}{\partial g} = \lambda_2(t)\big(p_x(t) - l\big)$, the pointwise Hamiltonian minimizer is

$$g^*(t) = \mathbf{1}\{p_x(t) \le l\} \quad \text{for a.e. } t \in [0, \tau_2]. \tag{20}$$

Since $H_t^e = f^{-1}(p_x(t))$ by definition, the threshold rule (20) is equivalently $g^*(t) = \mathbf{1}\{f(H_t^e) \le l\}$. For Phase 3, i.e., $t \in I_2$, note that $H_t^e = f^{-1}(p_x(t)) = \tau_2 - t$. For Phase 1, i.e., $t \in I_1$, we have $H_t^e = f^{-1}(p_x(t)) = (\tau_1 - t) - \frac{1}{\theta}\log\big(1 - \frac{\theta}{h}p_x(\tau_1)\big) = (\tau_1 - t) + f^{-1}(p_x(\tau_1)) = (\tau_1 - t) + C_1$.

Let $R(t) = p_x(t) - (1-p)p_b(t)$. On the second congestion period, the single-episode calculation gives $0 \le p_x < h/\theta$, $p_b \ge 0$, and $R \ge 0$. Across the gap, $p_b \ge 0$ and

$$R(t) = e^{-\mu(\sigma_2 - t)} R(\sigma_2) + (1-p) \int_t^{\sigma_2} e^{-\mu(s-t)} \nu(s) p_b(s)\, ds \ge 0.$$

Since $\dot{p}_x = \mu R$ on this interval, the explicit formula for p_x above implies $0 \le p_x(t) \le p_x(\sigma_2) < h/\theta$, for $t \in [\tau_1, \sigma_2]$. On the first congestion period, the expression for $p_x(t)$ is a convex combination of $p_x(\tau_1)$ and h/θ, with positive weight on $p_x(\tau_1)$. Hence $0 \le p_x(t) < h/\theta$ there as well. Moreover,

$$-\dot{R}(t) + \beta(t)R(t) = p\big(h - \theta p_x(t)\big) + \nu(t)p_x(t) \ge 0.$$

Backward integration from $R(\tau_1) \ge 0$ gives $R(t) \ge 0$ on the first congestion period. It follows that $A(t) = h - \theta p_x(t) + \mu R(t) > 0$ throughout $[0, \tau_2]$. Remark 1 therefore verifies global convexity and supplies the required nonnegative multipliers.

Overall, the state trajectory under the threshold rule $g^*(t) = \mathbf{1}\{p_x(t) \le l\} = \mathbf{1}\{f(H_t^e) \le l\}$ is feasible. The phasewise costate (p_x, p_b) satisfies the adjoint equations in each phase, matches continuously at phase boundaries, and satisfies transversality. Moreover, the control minimizes the phasewise Hamiltonian pointwise by (20). Theorem 3 therefore establishes optimality of g^*. □

A.3. Additional Two-Shock Analysis

In this section, we provide the proofs of Lemma 1 and Proposition 1.

Proof of Lemma 1 Write $y = (x, b)^\top$, let $v = g - g^*$, and set $g^\varepsilon = g^* + \varepsilon v$. The fluid vector field is locally Lipschitz in y and continuous across the congestion boundary. Differentiation of its integral equation gives the state variation $\xi = \partial_\varepsilon y^{g^\varepsilon}\big|_{\varepsilon = 0+}$, satisfying

$$\dot{\xi}(t) = A(t)\xi(t) + \begin{pmatrix} \lambda_2(t)v(t) \\ 0 \end{pmatrix}, \qquad \xi(0) = 0.$$

Here A is the state Jacobian along y^*. In the congested and uncongested regimes, respectively,

$$A_c(t) = \begin{pmatrix} -\theta & \mu - \theta \\ 0 & -((1-p)\mu + \nu(t)) \end{pmatrix}, \qquad A_u(t) = \begin{pmatrix} -\mu & 0 \\ (1-p)\mu & -\nu(t) \end{pmatrix}.$$

The state gradients of the holding cost are $r_c = (h, h)^\top$ and $r_u = (0, 0)^\top$.

The adjoint solves $-\dot{p} = A^\top p + r$, with $p(T) = 0$. Differentiating the objective and integrating by parts therefore yields

$$\begin{aligned} DJ(g^*; v) &= \int_0^T [r(t)^\top \xi(t) - l\lambda_2(t)v(t)]\, dt \\ &= \int_0^T [p(t)^\top (\dot{\xi}(t) - A(t)\xi(t)) - l\lambda_2(t)v(t)]\, dt = \int_0^T \lambda_2(t)(p_x(t) - l)v(t)\, dt. \end{aligned}$$

The boundary terms vanish because $\xi(0) = 0$ and $p(T) = 0$. □

Proof of Proposition 1 Along an optimal trajectory the Hamiltonian minimization condition gives $g^*(t) = 1$ when $p_x(t) < l$ and $g^*(t) = 0$ when $p_x(t) > l$. Values at equality do not contribute to the first-order penalty.

First, for t in either congestion period I_k, the all-accept trajectory from $(x^*(t), b^*(t))$ cannot clear before τ_k. Indeed, as long as both trajectories remain congested, their boarding equations are

identical and independent of the acceptance control. Their boarding states therefore coincide, and their treatment-population difference is

$$\tilde{x}(s) - x^*(s) = \int_t^s e^{-\theta(s-r)} \lambda_2(r)(1 - g^*(r))\, dr \geq 0.$$

An earlier clearing of the all-accept trajectory would contradict this inequality and the positivity of the optimal queue before τ_k. Consequently,

$$H_t(x^*(t), b^*(t)) \geq \tau_k - t, \quad t \in I_k. \tag{21}$$

For $t \in I_2$, if $\tau_2 - t > K$, both rules reject by (21). If $\tau_2 - t \leq K$, the optimal rule accepts throughout $[t, \tau_2)$ because its costate decreases over this interval. The all-accept and optimal trajectories then coincide up to clearing, so $H_t = \tau_2 - t$ and both rules accept. Thus I_2 contributes no mismatch penalty. After τ_2 both rules accept and the adjoint is zero.

For $t \in G_1$, strict uncongestion implies $H_t = 0$ and $\bar{g}^H(t) = 1$. The integrand of the mismatch penalty is therefore $\lambda_2(p_x(t) - l)^+$, giving the stated gap term.

For $t \in I_1$, put $u = \tau_1 - t$. If $p_x(t) = f(u + C_1) \leq l$, the optimal rule accepts from t until τ_1, so $H_t = u \leq K$ and the hitting-time rule also accepts. Hence disagreements can only occur when $p_x(t) > l$ and $H_t \leq K$. At any such time, $u + C_1 > K$ and $u \leq H_t \leq K$, which implies $K - C_1 < u \leq K$. Thus the transformed disagreement set is contained in $(K - C_1, K]$; it need not fill this interval. Integrating the nonnegative quantity $\lambda_2[f(u + C_1) - f(K)]$ over the full interval gives $\Delta^{\text{carry}} \leq \lambda_2 \int_{K-C_1}^{K} [f(u + C_1) - f(K)]\, du$.

With $v = u - (K - C_1)$, the upper bound equals

$$\frac{\lambda_2 h}{\theta} e^{-\theta K} \int_0^{C_1} (1 - e^{-\theta v})\, dv = \frac{\lambda_2 h}{\theta} e^{-\theta K} \left[C_1 - \frac{1 - e^{-\theta C_1}}{\theta} \right].$$

Using $1 - e^{-\theta v} \leq \theta v$ proves upper bound for Δ^{carry}.

Finally, $p_x(t)$ is increasing across G_1. To verify this directly, write $R = p_x - (1 - p)p_b$ and $\beta = (1 - p)\mu + \nu$. On the second congestion period, in backward time $s = \tau_2 - t$,

$$\frac{dR}{ds} + \beta(\tau_2 - s)R = \frac{h}{\theta} \left[\nu(\tau_2 - s)(1 - e^{-\theta s}) + p\theta e^{-\theta s} \right] > 0, \quad R(0) = 0.$$

Hence $R(\sigma_2) > 0$ and $p_b(\sigma_2) \geq 0$. Across the uncongested gap, $p_b(t) \geq 0$ and $\dot{R} = \mu R - (1 - p)\nu(t)p_b(t)$. Backward integration gives

$$R(t) = e^{-\mu(\sigma_2 - t)} R(\sigma_2) + (1 - p) \int_t^{\sigma_2} e^{-\mu(s-t)} \nu(s) p_b(s)\, ds > 0.$$

Thus $\dot{p}_x(t) = \mu R(t) > 0$ on G_1. If $\tau_2 - \sigma_2 \leq K$, then $p_x(\sigma_2) = f(\tau_2 - \sigma_2) \leq l$, so $p_x(t) \leq l$ throughout the gap and $\Delta^{\text{gap}} = 0$.

Appendix B: Asymptotic Optimality

We focus on the single surge setting for notational simplicity.

We first formalize the planning horizon T in (2). Given the initial state (x_0, b_0), define $T(x_0, b_0)$ as follows. Consider a fluid system that operates at full capacity and accepts all transfer requests, starting from (x_0, b_0). Its dynamics are $\dot{x}^o(t) = \lambda_1(t) + \lambda_2(t) - \mu\big(c(t) - b^o(t)\big) - \theta\big(x^o(t) + b^o(t) - c(t)\big)$, $\dot{b}^o(t) = (1 - p)\mu\big(c(t) - b^o(t)\big) - \nu(t)b^o(t)$. The planning horizon is the first time after the surge end at κ at which this reference trajectory clears:

$$T(x_0, b_0) = \inf\big\{ t \geq \kappa :\ x^o(t) + b^o(t) \leq c(t) \,\big|\, x^o(0) = x_0,\ b^o(0) = b_0 \big\}.$$

Assumption 1 ensures that $T<\infty$. The fluid dynamics are order preserving under the model's condition $\theta<\mu$: increasing admissions cannot decrease either state coordinate. Thus, every controlled fluid trajectory clears no later than the all-accept trajectory. Write $\bar{V}_T^*(x_0,b_0)$ for the optimal fluid value over $[0,T]$. Under these recovery conditions, extending a control by full acceptance after clearing incurs no additional cost, so $\bar{V}_T^*(x_0,b_0)=\bar{V}^*(x_0,b_0)$, where $\bar{V}^*$ is the clearing-horizon value in (5).

To establish asymptotic optimality, we consider a sequence of systems indexed by n, in which arrival rates, transfer request rates, and capacity scale linearly with n, while abandonment and service rates remain fixed: $\lambda_1^n(t)=n\lambda_1(t)$, $\lambda_2^n(t)=n\lambda_2(t)$, $c^n(t)=nc(t)$. We also scale the initial state as $(X^n(0),B^n(0))=(nx_0,nb_0)$. Let $Z^n(t)=\min\{X^n(t),c^n(t)-B^n(t)\}$ be the number in service and $G^n(t)\in\{0,1\}$ the transfer acceptance decision. Fix the same deterministic horizon $T=T(x_0,b_0)<\infty$ for all n.

Define the fluid-scaled processes $\bar{X}^n(t)=\frac{1}{n}X^n(t)$, $\bar{B}^n(t)=\frac{1}{n}B^n(t)$, $\bar{Z}^n(t)=\frac{1}{n}Z^n(t)$. For a policy π^n, the fluid-scaled cost is

$$\bar{V}^{n,\pi^n}(x_0,b_0)=\mathbb{E}\left[\int_0^{T(x_0,b_0)} h(\bar{X}^{n,\pi^n}(t)-\bar{Z}^{n,\pi^n}(t))\,dt+l\,\bar{L}^{n,\pi^n}(T(x_0,b_0))\right],$$

where $\bar{L}^{n,\pi^n}=L^{n,\pi^n}(t)/n$ is the scaled cumulative number of rejected transfers.

Recall that $H_t(x,b)$ is the fluid time to clear under full acceptance starting from (x,b) at time t, and $f(u)=\frac{h}{\theta}(1-e^{-\theta u})$, with $f(u)=hu$ when $\theta=0$. The following condition excludes initialization on the congestion boundary, where small stochastic excursions can prematurely trigger an irreversible switch.

Assumption 3 **(Strict initial congestion)**. *The initial state satisfies $x_0+b_0>c(0)$.*

By Theorem 1, an optimal fluid policy accepts all transfers from some time $\zeta^*\geq 0$ onward and rejects before that time. Write $g^*(t)=\mathbf{1}\{t\geq\zeta^*\}$ and (x^*,b^*,z^*) for its trajectory, extended by full acceptance after its clearing time τ^*. If $f(H_0(x_0,b_0))\leq l$, we take $\zeta^*=0$. Otherwise, $K:=f^{-1}(l)$ is well defined and $\zeta^*=\tau^*-K>0$.

Motivated by the fluid switching rule, define the stochastic look-ahead policy $\bar{\pi}^n$ to reject until

$$\zeta^n:=\inf\left\{t\geq 0: f\big(H_t(\bar{X}^n(t),\bar{B}^n(t))\big)\leq l\right\},\tag{22}$$

and accept thereafter. The infimum of the empty set is $+\infty$. For definiteness, a request that itself triggers the switch is rejected; subsequent requests are accepted. Thus, $G^n(t)=\mathbf{1}\{t>\zeta^n\}$.

Theorem 4. *Under Assumptions 1, for any sequence of admissible controls $\{\pi^n\}_{n\geq 1}$,*

$$\liminf_{n\to\infty}\bar{V}^{n,\pi^n}(x_0,b_0)\geq\bar{V}^*(x_0,b_0).\tag{23}$$

Moreover, if Assumption 3 also hold and $l>0$, for the sequence of systems operating under $\{\bar{\pi}^n\}_{n\geq 1}$,

$$\lim_{n\to\infty}\bar{V}^{n,\bar{\pi}^n}(x_0,b_0)=\bar{V}^*(x_0,b_0).\tag{24}$$

B.1. An Auxiliary Lemma

Lemma 2 **(Hitting-time continuity)**. *Under Assumption 1, the all-accept clearing time $H_t(x,b)$ is finite and jointly continuous in (t,x,b) on the feasible domain $\{(t,x,b): t\geq 0,\ x\geq 0,\ 0\leq b\leq c,\ x+b>c\}$.*

Proof. Fix a strictly congested initial state (x,b) at time t, and let $\tau = t + H_t(x,b)$ denote its absolute clearing time under full acceptance. Assumption 1 prevents clearing before κ and ensures finite clearing after $\max\{t,\kappa\}$. Moreover, at any boundary state $x+b=c$ and time $u \geq \kappa$, the drift of the total population satisfies $\lambda(u) - p\mu(c-b) - \nu(u)b \leq \lambda(u) - \min\{p\mu c, \nu(u)c\} \leq -\epsilon_2$. Thus, the reference trajectory crosses strictly into the uncongested region at τ.

By continuous dependence of the fluid solution on its starting time and state, nearby trajectories remain strictly congested up to $\tau - \varepsilon$ and are strictly uncongested at $\tau + \varepsilon$, for every sufficiently small $\varepsilon > 0$. Their absolute clearing times therefore converge to τ. Subtracting their starting times proves joint continuity of $H_t(x,b)$. $\square$

B.2. Proof of Theorem 4

Throughout, all processes are defined on a common probability space supporting the independent unit-rate Poisson processes driving the system.

We use a functional strong law of large numbers (FSLLN) for Poisson-driven systems with state-dependent Lipschitz rates (e.g., Mandelbaum et al. 1998): if the rates are locally bounded, Lipschitz in the fluid-scaled state, and of linear growth, then the fluid-scaled state process converges almost surely and uniformly on compact time intervals to the unique solution of the corresponding deterministic integral equation. We apply this result on time intervals where the acceptance rule is constant (all reject or all accept).

Proof of the lower bound (23). Fix any sequence of admissible policies $\{\pi^n\}$, and denote the associated admission decisions by $G^n(t) \in \{0,1\}$. Write $Y^n(t) = (\bar{X}^n(t), \bar{B}^n(t))$, $y = (x,b)$, $z(t,y) = \min\{x, c(t)-b\}$ and $q(t,y) = x - z(t,y) = (x+b-c(t))^+$. The deterministic fluid drift is

$$F(t,y,g) = \begin{pmatrix} \lambda_1(t) + \lambda_2(t)g - \mu z(t,x,b) - \theta q(t,x,b) \\ (1-p)\mu z(t,x,b) - \nu(t)b \end{pmatrix}.$$

Because z and q are Lipschitz in y and ν is bounded, there is a finite constant L_F, independent of n and the policy, such that $\|F(t,y,g) - F(t,\widetilde{y},g)\|_1 \leq L_F \|y - \widetilde{y}\|_1$.

Compensating the arrival, service, abandonment, and boarding-completion counts gives

$$Y^n(t) = y_0 + \int_0^t F(s, Y^n(s), G^n(s))\, ds + M^n(t), \tag{25}$$

where M^n is a two-dimensional square-integrable martingale.

We first establish a bound on the martingale that is uniform over admissible policies. Each component of M^n has jumps of magnitude $1/n$. The expected total population is bounded by the initial population plus expected arrivals:

$$\mathbb{E}[X^n(t) + B^n(t)] \leq n\left[x_0 + b_0 + \int_0^T (\lambda_1(s) + \lambda_2(s))\, ds\right] =: nK_0.$$

Since the arrival intensities are bounded by $n(\lambda_1+\lambda_2)$, and Z^n, Q^n, B^n have expectations of order n uniformly over policies, the expected integrated intensities of all transition counts on $[0,T]$ are bounded by a constant times n. Therefore, for a finite constant K_M independent of the policy,

$$\mathbb{E}\langle M_X^n\rangle_T + \mathbb{E}\langle M_B^n\rangle_T \leq \frac{K_M}{n}.$$

Doob's L^2 inequality, followed by Cauchy–Schwarz, yields a finite constant K_1 such that

$$\mathbb{E}\left[\sup_{0\leq t\leq T} \|M^n(t)\|_1\right] \leq \frac{K_1}{\sqrt{n}}. \tag{26}$$

For each realization of G^n, construct a fluid trajectory y_f^n using exactly that control path:

$$y_f^n(t) = y_0 + \int_0^t F(s, y_f^n(s), G^n(s))\, ds. \tag{27}$$

The trajectory is random through G^n, but for each realization it is a feasible work-conserving fluid trajectory driven by a measurable control in $[0,1]$.

Subtracting (27) from (25) gives $\|Y^n(t) - y_f^n(t)\|_1 \le \|M^n(t)\|_1 + L_F \int_0^t \|Y^n(s) - y_f^n(s)\|_1\, ds$. Gronwall's inequality and (26) imply

$$\mathbb{E}\left[\sup_{0\le t\le T} \|Y^n(t) - y_f^n(t)\|_1\right] \le \frac{K_1 e^{L_F T}}{\sqrt{n}}.$$

For every realization, the function $t \mapsto G^n(t)$ is a feasible fluid control. Hence

$$\int_0^T \big[hq(t, y_f^n(t)) + l\lambda_2(t)(1 - G^n(t))\big]\, dt \ge \bar{V}_T^*(x_0, b_0) \qquad \text{almost surely.}$$

The rejection count has compensator $\int_0^T n\lambda_2(t)(1 - G^n(t))dt$, so $\mathbb{E}\,[L^n(T)/n] = \mathbb{E}\int_0^T \lambda_2(t)(1 - G^n(t))dt$. Consequently, the fluid-scaled stochastic cost satisfies

$$\begin{aligned}
\bar{V}^{n,\pi^n}(x_0, b_0) &= \mathbb{E}\int_0^T \big[hq(t, Y^n(t)) + l\lambda_2(t)(1 - G^n(t))\big]\, dt \\
&\ge \mathbb{E}\left[\int_0^T \big[hq(t, y_f^n(t)) + l\lambda_2(t)(1 - G^n(t))\big]\, dt\right] - h\int_0^T \mathbb{E}|q(t, Y^n(t)) - q(t, y_f^n(t))|\, dt \\
&\ge \bar{V}_T^*(x_0, b_0) - \frac{hTK_1 e^{L_F T}}{\sqrt{n}}.
\end{aligned}$$

The error bound is uniform over admissible policies. Taking lim inf yields (23).

Proof of (24). Now consider the policy $\{\bar{\pi}^n\}$ defined by (22). Let $(x^*(t), b^*(t), z^*(t), g^*(t))$ be the optimal fluid solution, where $g^*(t) = \mathbf{1}\{t \ge \zeta^*\}$. We prove that under $\bar{\pi}^n$,

$$\sup_{0\le t\le T} \big|(\bar{X}^n(t), \bar{B}^n(t)) - (x^*(t), b^*(t))\big| \xrightarrow[n\to\infty]{a.s.} 0, \tag{28}$$

and then show convergence of costs.

Step 1: FSLLN on intervals where the acceptance decision is constant. For any deterministic $s \in [0,T]$, consider the system with $G^n(t) = 0$ throughout $[0,s]$. Its arrival and departure rates are Lipschitz functions of the fluid-scaled state, with linear growth, because $\bar{Z}^n(t) = \min\{\bar{X}^n(t), c(t) - \bar{B}^n(t)\}$ is Lipschitz in the state and all primitive rates are bounded. The FSLLN therefore gives almost sure uniform convergence to the fluid trajectory driven by the same reject decision. The same argument applies under the constant accept decision.

Step 2: Convergence of the switching times $\zeta^n \to \zeta^*$. Write $\Psi^*(t) = f(H_t(x^*(t), b^*(t)))$ and $\Psi^n(t) = f(H_t(\bar{X}^n(t), \bar{B}^n(t)))$. The switching times satisfy $\zeta^* = \inf\{t \ge 0 : \Psi^*(t) \le l\}$ and $\zeta^n = \inf\{t \ge 0 : \Psi^n(t) \le l\}$. If $\zeta^* = 0$, then $f(H_0(x_0, b_0)) \le l$ and the exact initialization implies $\zeta^n = 0$ for every n.

Now suppose $\zeta^* > 0$, and recall $K = f^{-1}(l) > 0$ and $\tau^* - \zeta^* = K$. Fix $\varepsilon > 0$ sufficiently small. Before ζ^*, accepting all transfers cannot clear the queue earlier than the optimal trajectory. After ζ^*, the optimal policy accepts all transfers. Therefore,

$$\inf_{0\le t\le \zeta^* - \varepsilon} \Psi^*(t) \ge f(K + \varepsilon) > l, \quad \Psi^*(\zeta^* + \varepsilon) = f(K - \varepsilon) < l. \tag{29}$$

Let (x^R, b^R) denote the fluid trajectory that rejects all transfers, and let τ_R be its clearing time. When $\zeta^* > 0$, this trajectory agrees with the optimal trajectory up to ζ^* and is still strictly congested there, so $\zeta^* < \tau_R$. Let $(\bar{X}^{n,R}, \bar{B}^{n,R})$ denote the scaled stochastic system that always rejects. By Assumption 3 and $\zeta^* < \tau_R$, there exists $\delta > 0$ such that $\zeta^* + \delta < \tau_R$ and the all-reject trajectory is strictly congested throughout $[0, \zeta^* + \delta]$. Lemma 2 therefore ensures that $(t, x, b) \mapsto H_t(x, b)$ is jointly continuous on a neighborhood, relative to the feasible domain, of this trajectory segment. Then, by Step 1, $\sup_{0 \le t \le \zeta^* + \delta} \left| f(H_t(\bar{X}^{n,R}(t), \bar{B}^{n,R}(t))) - f(H_t(x^R(t), b^R(t))) \right| \longrightarrow 0$ almost surely. On $[0, \zeta^* - \varepsilon]$, the all-reject and optimal fluid trajectories agree. The first inequality in (29) therefore implies that the stochastic policy cannot switch on this interval for sufficiently large n. Consequently, $\liminf_n \zeta^n \ge \zeta^* - \varepsilon$ almost surely.

For the reverse bound, continuing to reject after ζ^* gives a state no larger than accepting. The all-accept clearing time is nondecreasing in the initial state, so

$$f(H_{\zeta^* + \varepsilon}(x^R(\zeta^* + \varepsilon), b^R(\zeta^* + \varepsilon))) \le f(K - \varepsilon) < l.$$

The same uniform convergence implies that the stochastic all-reject statistic is below l at $\zeta^* + \varepsilon$ for sufficiently large n. If the look-ahead policy had not switched by then, its state would still equal the all-reject state, contradicting (22). Hence $\limsup_n \zeta^n \le \zeta^* + \varepsilon$ almost surely. Letting $\varepsilon \downarrow 0$ along a countable sequence proves $\zeta^n \to \zeta^*$ almost surely.

Step 3: Stitching the fluid limits across the (asymptotically matching) switching times. Define $\tau_-^n := \zeta^n \wedge \zeta^*$ and $\tau_+^n := \zeta^n \vee \zeta^*$. On $[0, \tau_-^n]$, both the stochastic policy and the fluid optimal policy reject transfers. The FSLLN in Step 1 implies $\sup_{0 \le t \le \tau_-^n} \left| (\bar{X}^n(t), \bar{B}^n(t)) - (x^*(t), b^*(t)) \right| \xrightarrow[n \to \infty]{a.s.} 0$. Next, we handle the transition. By Step 2, $\tau_+^n - \tau_-^n \to 0$ a.s. Using the local boundedness of the transition rates, the state cannot change macroscopically during this vanishing interval. Thus,

$$(\bar{X}^n(\tau_+^n), \bar{B}^n(\tau_+^n)) \xrightarrow[n \to \infty]{a.s.} (x^*(\zeta^*), b^*(\zeta^*)).$$

On the remaining interval $[\tau_+^n, T]$, both policies accept transfers. Let $(\hat{x}^n(t), \hat{b}^n(t))$ denote the unique solution to the Accept fluid ODE starting at time τ_+^n with initial state $(\bar{X}^n(\tau_+^n), \bar{B}^n(\tau_+^n))$. We decompose the error using the triangle inequality:

$$|(\bar{X}^n(t), \bar{B}^n(t)) - (x^*(t), b^*(t))| \le \underbrace{|(\bar{X}^n(t), \bar{B}^n(t)) - (\hat{x}^n(t), \hat{b}^n(t))|}_{\text{(A)}} + \underbrace{|(\hat{x}^n(t), \hat{b}^n(t)) - (x^*(t), b^*(t))|}_{\text{(B)}}.$$

Term (A) vanishes almost surely by applying the FSLLN (as in Step 1) to the process starting at τ_+^n. Term (B) vanishes by the continuous dependence of ODE solutions on initial state. Then,

$$\sup_{\tau_+^n \le t \le T} \left| (\bar{X}^n(t), \bar{B}^n(t)) - (x^*(t), b^*(t)) \right| \xrightarrow[n \to \infty]{a.s.} 0.$$

Combining the convergence on $[0, \tau_-^n]$ and $[\tau_+^n, T]$ proves (28).

Step 4: Convergence of costs. First, the function $\bar{Z}^n(t) = \min\{\bar{X}^n(t), c(t) - \bar{B}^n(t)\}$ is continuous in the state $(\bar{X}^n, \bar{B}^n)$. Given the uniform convergence established in (28), we have $\sup_{0 \le t \le T} |\bar{Z}^n(t) - z^*(t)| \xrightarrow[n \to \infty]{a.s.} 0$. Next, the rejection indicator under $\bar{\pi}^n$ is $1 - G^n(t) = \mathbf{1}\{t < \zeta^n\}$, so

$$\int_0^T \lambda_2(t)(1 - G^n(t))dt = \int_0^{\zeta^n} \lambda_2(t)\, dt \xrightarrow[n \to \infty]{a.s.} \int_0^{\zeta^*} \lambda_2(t)\, dt,$$

by $\zeta^n \to \zeta^*$ and boundedness of $\lambda_2(\cdot)$.

Finally, since the fluid-scaled states are uniformly integrable (due to being dominated by the intial state and the square-integrable arrival process over the finite horizon $[0,T]$), we obtain

$$\lim_{n\to\infty} \bar{V}^{n,\bar{\pi}^n}(x_0,b_0) = \int_0^T h\big(x^*(t)-z^*(t)\big)dt + \int_0^T l\,\lambda_2(t)\big(1-g^*(t)\big)dt = \bar{V}^*(x_0,b_0),$$

establishing (24). □

Appendix C: Case Study

C.1. Data Processing

For transfer patients data, we exclude all ED-to-inpatient and inpatient-to-inpatient transfer requests, as these do not constitute ED arrivals. All ED-to-ED transfer requests are included regardless of whether the requests are accepted or declined (these account for 36% of all documented transfer requests). Together with non-transfer ED patients (external arrivals), they constitute our study cohort. We further exclude patients requesting inpatient services outside of general medicine, oncology, cardiology, surgery, neurology, or orthopedics, which accounts for 3.20% of ED patients. We include all admitted inpatients when computing hourly inpatient occupancy for each of the medical services listed above. In addition, we control for local hourly weather conditions (temperature, precipitation, snowfall, and wind speed) in all predictive analyses. Finally, to ensure the accuracy of occupancy and census measures, we exclude observations from the initial and terminal months of the data. The resulting final cohort covers April 2022 through November 2024 and consists of 311,739 ED encounters. This case study was conducted with relevant Institutional Review Board approval; the need for informed consent was waived.

C.2. Estimate Processing Rate

We model the per-patient processing rate from ED boarding to inpatient admission as piecewise constant over pre-specified 8-hour intervals. Specifically, conditional on still boarding at time $t\in[t_k,t_{k+1})$, each patient completes boarding at instantaneous rate ν_k, regardless of their elapsed boarding time. For each interval k, we include all patients whose boarding periods overlap $[t_k,t_{k+1})$. Let b_i and d_i denote patient i's boarding start and end times, respectively, and let $\mathcal{I}_k$ denote the set of patients included in interval k. Define patient i's boarding exposure within the interval as $B_{ik}=\min(d_i,t_{k+1})-\max(b_i,t_k)$, and let $\delta_{ik}=\mathbf{1}\{d_i\in[t_k,t_{k+1})\}$, indicating whether the patient completes boarding within the interval. Patients who remain in boarding status at the end of the interval are treated as right-censored for that interval. Conditional on the patient still boarding at $\max(b_i,t_k)$, the likelihood contribution is $\nu_k e^{-\nu_k B_{ik}}$ for a completed observation and $e^{-\nu_k B_{ik}}$ for a right-censored observation. Combining these contributions yields $L_k(\nu_k)=\prod_{i\in\mathcal{I}_k}\nu_k^{\delta_{ik}}e^{-\nu_k B_{ik}}$. Provided that total boarding exposure is positive, the maximum likelihood estimator is $\widehat{\nu}_k=\frac{\sum_{i\in\mathcal{I}_k}\delta_{ik}}{\sum_{i\in\mathcal{I}_k}B_{ik}}$. Thus, the estimated processing rate equals the number of boarding completions divided by the total boarding patient-hours within the interval. We estimate the rate separately for each interval and set $\nu(t)=\widehat{\nu}_k$ for $t\in[t_k,t_{k+1})$.

Table 3 summarizes the arrival counts (per two-hour interval), boarding delays, and estimated processing rates. On average, there are 26.64 arrivals every two hours, the mean boarding delay is 17.88 hours, and the average processing rate is 0.06.

Statistic	Arrival Count (per 2 hours)	Boarding Delay (hours)	Estimated Processing Rate
Count	11700	65582	2925
Mean	26.64	17.88	0.06
Standard deviation	13.07	15.42	0.04
Min	1	0.02	0.00
25%ile	15	5.08	0.03
50%ile	27	14.78	0.06
75%ile	37	25.12	0.08
Max	67	155.03	0.30

Table 3 Summary statistics

C.3. Characterize Arrival Pattern

To characterize the temporal spacing of high-demand periods in the empirical data, we define a high-arrival interval as a two-hour interval in which the arrival count exceeds its empirical 90th percentile, and group consecutive high-arrival intervals into a single high-arrival episode. We identify 875 such episodes over the study period. The median episode lasts 2 hours, and the median gap between consecutive episodes is 18 hours. Moreover, 25.9% of episodes are followed by another high-arrival episode within 6 hours and 75.8% within 24 hours. Thus, high-arrival episodes in the empirical data are not necessarily separated by long recovery periods.

C.4. Additional Results of Case Study

In Section 6.3, we simulate the system under hospital's current policy with thresholds $\bar{b}$. Figure 9(a) summarizes the results for $\delta = 1$ and $\gamma = 0.1$. As $\bar{b}$ increases, average waiting time rises, while the fraction of rejected transfer requests declines, reflecting the congestion–access trade-off. We next evaluate the look-ahead policy (13) over the parameter grid in (14). Figure 9(b) shows the resulting trade-off between congestion and access for $\delta = 1$ and $\gamma = 0.1$. As l/h increases, the policy places greater weight on access, rejects fewer transfer requests, and tolerates longer waiting times. Figure 9(c) presents the analogous trade-off between boarding census and transfer rejection rate. Results for all decision window sizes, $\delta \in \{1, 2, 4, 8, 12\}$ hours, and $\gamma \in \{0.033, 0.1, 0.2\}$, are qualitatively similar and are reported in Figures 10. As discussed in Section 6.3, these patterns are less pronounced when $\gamma = 3.3\%$, as shown in the top row of Figure 10, because transfer patients constitute a smaller share of arrivals.

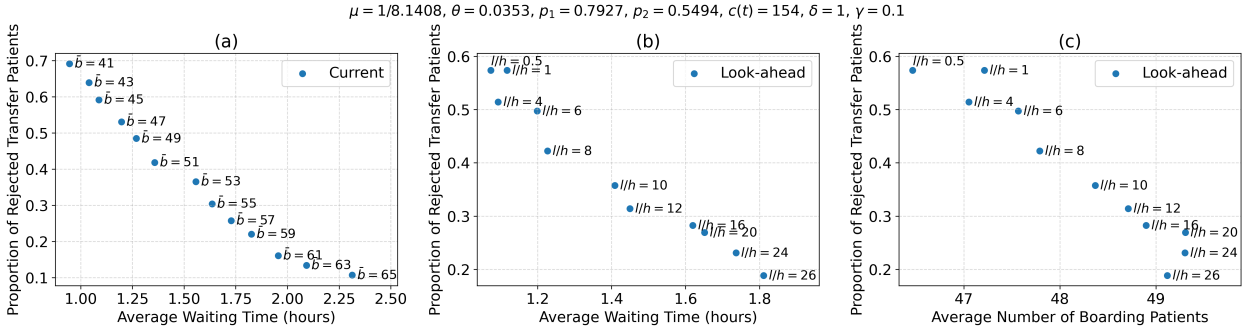


Figure 9 Simulation results under hospital's current policy and look-ahead policy with $\delta = 1$ and $\gamma = 0.1$.

C.5. Robustness to an Expanded Forecasting Specification

As a robustness check, we expand the forecasting specification in Section 6.2 to incorporate additional contemporaneous operational information. Specifically, for arrival-rate prediction, we additionally include the ED census at the beginning of each interval. For processing-rate prediction, we further include the current boarding census and service-level inpatient occupancy measures. These

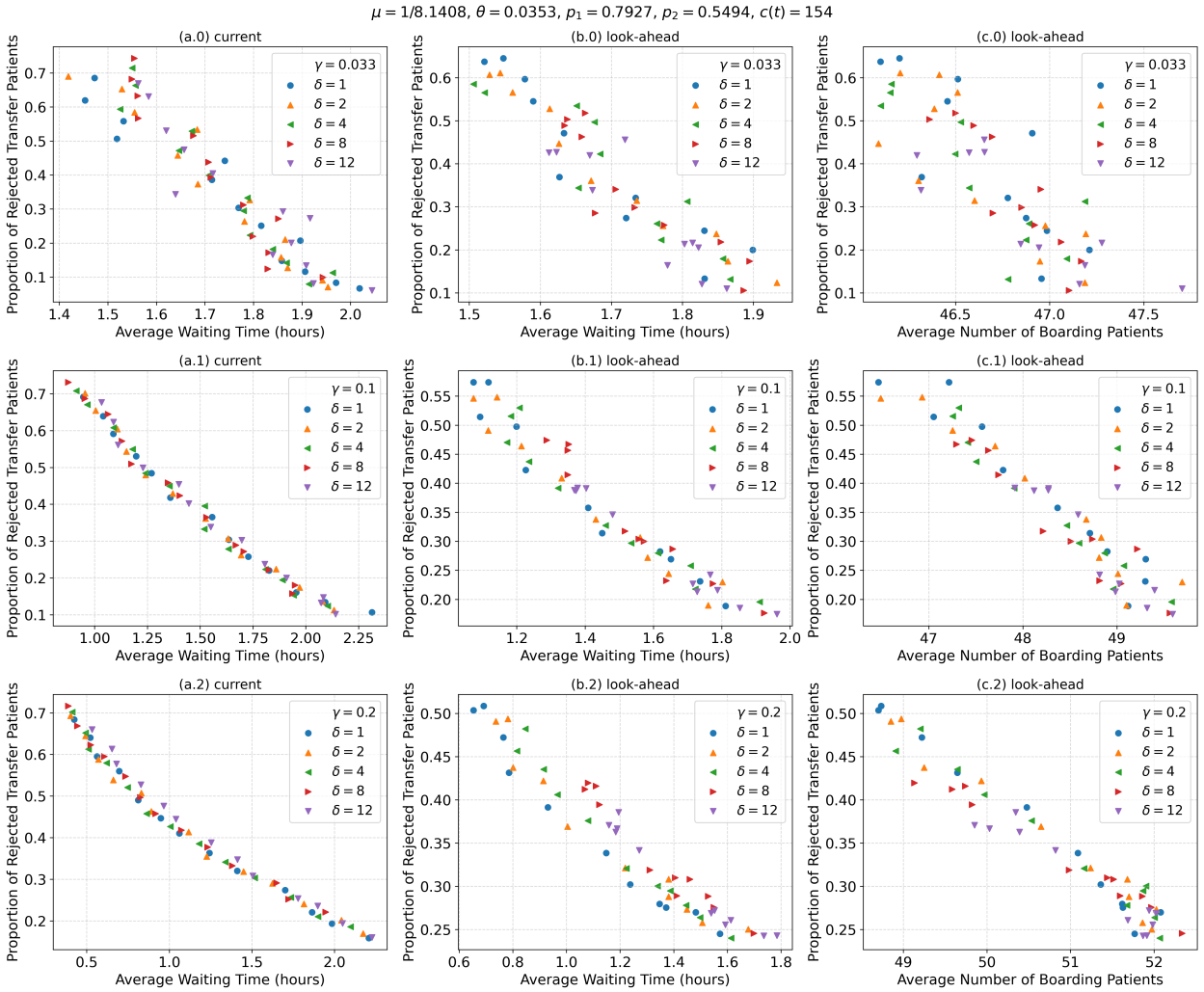


Figure 10 Trade-offs between the proportion of rejected transfers and average waiting time or boarding census.

variables provide a richer characterization of the system state. In this exercise, we use their realized values for the intervals being predicted; an operational implementation would require additional models to forecast their future values. Accordingly, this exercise assesses sensitivity to a richer information set.

Table 4 reports the predictive accuracy of the expanded arrival and processing rate models. Test-set RMSEs are similar to those under the main specification. For linear regression, R^2 remains at 81.7% for arrivals and increases from 69.1% to 75.1% for processing rates.

Prediction Target	**Dataset**	**Linear Regression**	**XGBoost**
Arrival count	Train	5.61	4.23
	Test	5.69	5.67
Processing rate	Train	0.02	0.01
	Test	0.02	0.02

Table 4 RMSE values for the arrival and processing rate prediction models.

Using forecasts from the expanded XGBoost models, with realized values of the additional operational covariates, we repeat the simulation study described in Section 6.3. All other simulation settings remain unchanged.

Figure 11 compares the hospital's current policy with the look-ahead policy under the main and expanded forecasting specifications. Both look-ahead specifications achieve a more favorable congestion-access trade-off than the hospital's policy. Their performance is similar, consistent with their comparable predictive accuracy.

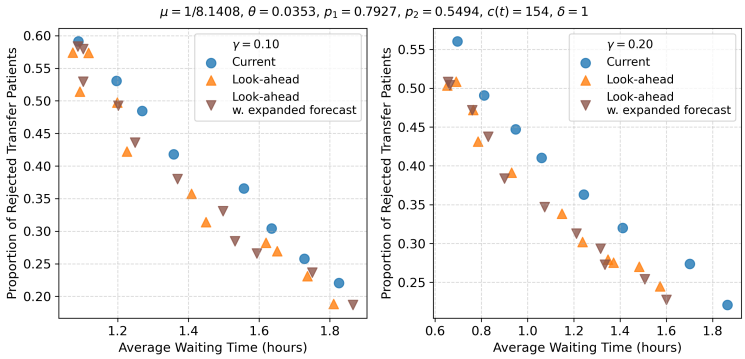


Figure 11 Comparison between the look-ahead policies and the hospital's current policy

Appendix D: Additional Numerical Experiments

This section reports the numerical computation of H_t and additional numerical experiments that supplement the results presented in Section 5.

Numerical computation of H_t. The uncontrolled system in (6) is solved numerically forward from $(\widetilde{x}(t), \widetilde{b}(t)) = (x, b)$ under $g(u) \equiv 1$. For intuition, under a forward-Euler discretization with step size Δ, given $(\widetilde{x}_j, \widetilde{b}_j)$ at time u_j, we set $\widetilde{z}_j = \min\{\widetilde{x}_j, c(u_j) - \widetilde{b}_j\}$, $\widetilde{q}_j = \widetilde{x}_j - \widetilde{z}_j$, and update $\widetilde{x}_{j+1} = \widetilde{x}_j + \Delta\left[\lambda(u_j) - \mu\widetilde{z}_j - \theta\widetilde{q}_j\right]$, $\widetilde{b}_{j+1} = \widetilde{b}_j + \Delta\left[(1-p)\mu\widetilde{z}_j - \nu(u_j)\widetilde{b}_j\right]$. The iterations continue until the queue clears, and the corresponding elapsed time gives $H_t(x, b)$. In our implementation, we use the `solve_ivp` routine from `SciPy` scientific computing library in Python with the `LSODA` method and adaptive time stepping rather than a fixed-step Euler scheme, together with event detection to identify the first queue-clearing time.

Benchmark policy performance. Figure 12 presents the trade-off curves of benchmark policies with $\eta \in \{0, 0.5, 1\}$ showing that, when Γ is appropriately tuned, the performance of the benchmark policy is largely insensitive to the choice of η.

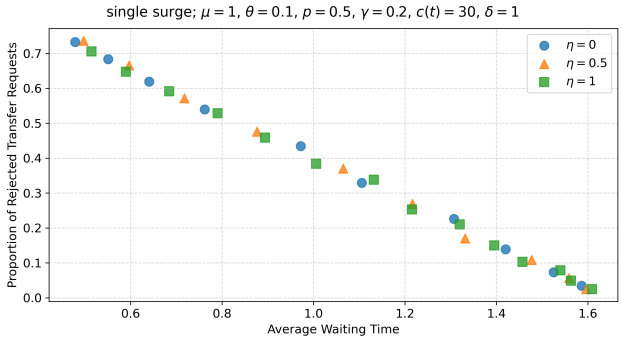


Figure 12 Trade-off under benchmark policy with different η.

Time-varying processing rate $\nu(t)$. We next consider a time-varying processing rate $\nu(t)$ with a congestion surge, where $\nu(t) = 0.7$ for $0 \le t \le 50$, $\nu(t) = 0.3$ for $50 < t \le 100$, $\nu(t) = 0.7$ for

$100 < t \le T$, and $\nu(t) = 0$ otherwise. The left panel of Figure 13 illustrates the resulting profile of the processing rate $\nu(t)$. The arrival rate is set to $\lambda(t) = 15$ throughout the horizon.

To compare the performance of the benchmark and look-ahead policies, the right panel of Figure 13 depicts the trade-off between patient waiting time and the proportion of rejected transfer patients. Across both panels (corresponding to review intervals $\delta = 1$ and $\delta = 5$) the look-ahead policy consistently dominates the benchmark by achieving lower rejection rates for comparable or shorter waiting times. These results further highlight the value of incorporating predictive information into proactive admission control decisions.

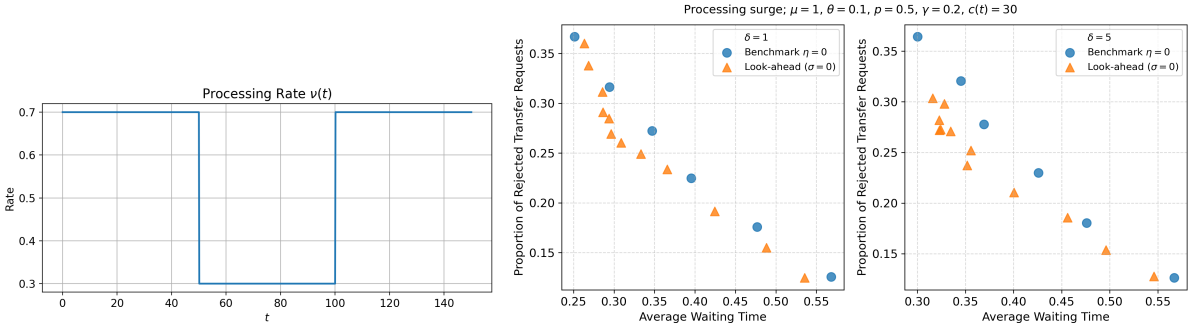


Figure 13 Comparison between look-ahead v.s. benchmark under surge processing rate.

Prediction errors. In Section 5.2, Scenarios I–V examine all combinations of a single congestion shock, two congestion shocks, $\sigma \in \{1, 2\}$, and $\delta \in \{1, 5\}$. Complete results for all configurations are available at `https://tinyurl.com/e6m37vw2`. The qualitative insights reported in Section 5.2 are consistent across all experiments.

Appendix E: Extension to Multiple Patient Classes

The main model aggregates patients receiving ED treatment into a single class. This appendix considers two patient classes that differ in their clinical and cost parameters and are served under a preemptive-priority discipline. We develop a class-specific look-ahead admission heuristic and compare its performance with a benchmark based on the fluid control.

E.1. Two-Class Model

Stochastic model. Let $\mathcal{K} = \{H, L\}$ denote the patient classes, with class H receiving preemptive priority over class L. Patients arrive through external arrivals ($i = 1$) and transfer requests ($i = 2$), with aggregate rates $\lambda_i(t)$. Let α_{ik} denote the proportion of stream i belonging to class k, where $\alpha_{iH} + \alpha_{iL} = 1$. The class-specific arrival rates are $\lambda_{ik}(t) = \alpha_{ik}\lambda_i(t)$, $i \in \{1, 2\}$ and $k \in \mathcal{K}$. Thus, external arrivals and transfer requests may have different class compositions.

A class-k patient has an exponential ED service time with rate μ_k and an exponential patience time with rate θ_k. Following ED treatment, the patient is discharged with probability p_k and requires inpatient admission with probability $1 - p_k$. Patients requiring inpatient admission enter a common boarding pool, with the time-varying processing rate $\nu(t)$, and continue to occupy ED servers until their inpatient transfers are completed.

Let $X_k(t)$, $Z_k(t)$, and $Q_k(t)$ denote the number of class-k patients who have not completed ED treatment, are receiving treatment, and are waiting, respectively, so that $X_k(t) = Z_k(t) + Q_k(t)$. Let $B(t)$ denote the number of patients boarding and define $C(t) = [c(t) - B(t)]^+$. Under

work conservation and preemptive priority, $Z_H(t) = \min\{X_H(t), C(t)\},\ Z_L(t) = \min\{X_L(t), [C(t) - Z_H(t)]^+\}$. Patients are served FCFS within each class, and boarding patients cannot be preempted.

Let $G_k(t) \in \{0,1\}$ indicate whether a class-k transfer request arriving at time t is accepted. Let $A_{ik},\ S_{d,k},\ S_{b,k},\ D_k$, and R be independent rate-one Poisson processes representing class-specific arrivals, service completions, abandonments, and inpatient transfer completions, respectively. The stochastic dynamics are $X_k(t) = X_k(0) + A_{1k}\left(\int_0^t \lambda_{1k}(u)\,du\right) + A_{2k}\left(\int_0^t \lambda_{2k}(u)G_k(u)\,du\right) - S_{d,k}\left(\int_0^t p_k\mu_k Z_k(u)\,du\right) - S_{b,k}\left(\int_0^t (1-p_k)\mu_k Z_k(u)\,du\right) - D_k\left(\int_0^t \theta_k Q_k(u)\,du\right),\ k \in \mathcal{K}$, and $B(t) = B(0) + \sum_{k\in\mathcal{K}} S_{b,k}\left(\int_0^t (1-p_k)\mu_k Z_k(u)\,du\right) - R\left(\int_0^t \nu(u)B(u)\,du\right)$.

Fluid model and optimal control problem. Let x_k, z_k, q_k, and b denote the corresponding fluid quantities, and let $g_k(t) \in [0,1]$ be the fraction of transfer fluid accepted. The fluid dynamics satisfy

$$\dot{x}_k(t) = \lambda_{1k}(t) + g_k(t)\lambda_{2k}(t) - \mu_k z_k(t) - \theta_k q_k(t),\ k \in \mathcal{K},\ \dot{b}(t) = \sum_{k\in\mathcal{K}} (1-p_k)\mu_k z_k(t) - \nu(t)b(t), \quad (30)$$

where $q_k(t) = x_k(t) - z_k(t), z_H(t) = \min\{x_H(t), [c(t) - b(t)]^+\}, z_L(t) = \min\{x_L(t), [c(t) - b(t) - z_H(t)]^+\}$.

To define the congestion-episode horizon, let (x_H^o, x_L^o, b^o) denote the all-admit fluid trajectory, obtained by setting $g_H = g_L = 1$. For initial state $(\mathbf{x}_0, b_0)$, where $\mathbf{x}_0 = (x_{H,0}, x_{L,0})$, define $T(\mathbf{x}_0, b_0) = \inf\{t \geq \kappa : x_H^o(t) + x_L^o(t) + b^o(t) \leq c(t)\}$, where κ is the end of the congestion shock. This horizon is determined by the reference trajectory and is fixed across admission policies. Given class-specific holding-cost rates h_k and rejection costs l_k, the fluid-control problem is

$$\overline{V}^*(\mathbf{x}_0, b_0) = \inf_{g_H, g_L} \int_0^{T(\mathbf{x}_0,b_0)} \sum_{k\in\mathcal{K}} \left[h_k q_k(t) + l_k \lambda_{2k}(t)(1 - g_k(t))\right] dt, \quad (31)$$

subject to (30) and $g_k(t) \in [0,1]$.

E.2. Look-Ahead Heuristic

At each review epoch t, we initialize the fluid model at the observed state and propagate it using forecasts available at that epoch, assuming that all subsequent transfer requests are accepted. Denote the resulting reference trajectory by $\left(\widehat{x}_H^o(s \mid t), \widehat{x}_L^o(s \mid t), \widehat{b}^o(s \mid t)\right),\ s \geq t$. Define the predicted clearing times $\widehat{H}_{H,t} = \inf\left\{u \geq 0 : \widehat{x}_H^o(t+u \mid t) + \widehat{b}^o(t+u \mid t) \leq c(t+u)\right\},\ \widehat{H}_{L,t} = \inf\left\{u \geq 0 : \widehat{x}_H^o(t+u \mid t) + \widehat{x}_L^o(t+u \mid t) + \widehat{b}^o(t+u \mid t) \leq c(t+u)\right\}$. The first is the predicted time until sufficient capacity is available to serve all class-H patients; the second is the predicted time until sufficient capacity is available to serve all patients. Since the second condition implies the first, $\widehat{H}_{H,t} \leq \widehat{H}_{L,t}$. For $k \in \mathcal{K}$, define $F_k(u) = h_k \int_0^u e^{-\theta_k s}\,ds = \dfrac{h_k}{\theta_k}(1 - e^{-\theta_k u})$, if $\theta_k > 0$, and $F_k(u) = h_k u$, if $\theta_k = 0$. This function measures the cumulative holding cost of an initially unit mass of class-k fluid that remains waiting until time u unless it abandons.

We use these functions to construct the congestion indices $\widehat{C}_L(t) = F_L(\widehat{H}_{L,t}),\ \widehat{C}_H(t) = F_H(\widehat{H}_{H,t}) + e^{-\theta_H \widehat{H}_{H,t}} F_L(\widehat{H}_{L,t} - \widehat{H}_{H,t})$. The class-$L$ index uses the total clearing time to approximate the congestion cost associated with a low-priority admission. The class-H index has two components: a term for its own waiting cost before high-priority congestion clears, and a correction for the lower-priority service it may displace. The latter uses the interval between the two clearing times, weighted by the probability of surviving the high-priority waiting period without abandoning. This cross-class correction is a heuristic approximation rather than a general characterization of the fluid marginal cost. It is most naturally motivated by a nested congestion episode in which high-priority waiting clears before total waiting.

At each review epoch, the heuristic accepts class-k transfer requests if their rejection cost is at least as large as the corresponding congestion index: $G_k(t) = \mathbf{1}\{\widehat{C}_k(t) \le l_k\},\ k \in \mathcal{K}$. The acceptance indicators remain fixed until the next review epoch.

E.3. Numerical Evaluation

We compare the two-class look-ahead heuristic with a fluid-optimization benchmark based on (31). Both policies are evaluated in the two-surge stochastic system described in Section 5, with review interval $\delta = 1$. We set $\alpha_{1H} = 0.3,\ \alpha_{2H} = 0.5,\ \mu_H = 0.6,\ \mu_L = 0.9,\ p_H = 0.6,\ p_L = 0.8,\ \theta_H = \theta_L = 0.1$. Thus, transfer requests contain a larger proportion of high-priority patients than external arrivals, and high-priority patients have longer mean treatment times and a greater probability of requiring inpatient admission. We fix $h_L = 1$ and $h_H = 2$, and vary the rejection costs over $l_L \in \{0.5, 1, 2, \ldots, 9, 9.9\},\ l_H \in \{1, 2, 4, 6, 8, 10, 12, 14, 16, 18, 19\}$. Restricting attention to $l_H \ge l_L$ yields 92 parameter combinations. For each combination, we perform 200 replications per policy.

Figure 14 reports mean total costs and 95% confidence intervals, with parameter combinations ordered by the mean cost of the fluid-optimization benchmark. The look-ahead heuristic achieves costs close to those of the benchmark across the tested settings. These results provide numerical support for using class-specific clearing-time indices in a system with heterogeneous patients and service priority.

The heuristic also requires substantially less computation. At each review epoch, it propagates a single fluid reference trajectory and performs two scalar cost comparisons. The benchmark instead solves a fluid optimization problem at every review epoch. Its computational requirements depend on the current state and the remaining planning horizon, and some updates may fail to produce a solution within the time budget. Evaluating all 92 parameter combinations, with 200 replications per combination and 40 runs in parallel, has required approximately 28 minutes of wall-clock time for the look-ahead policy, compared with at least 1,280 minutes for the fluid-optimization benchmark.

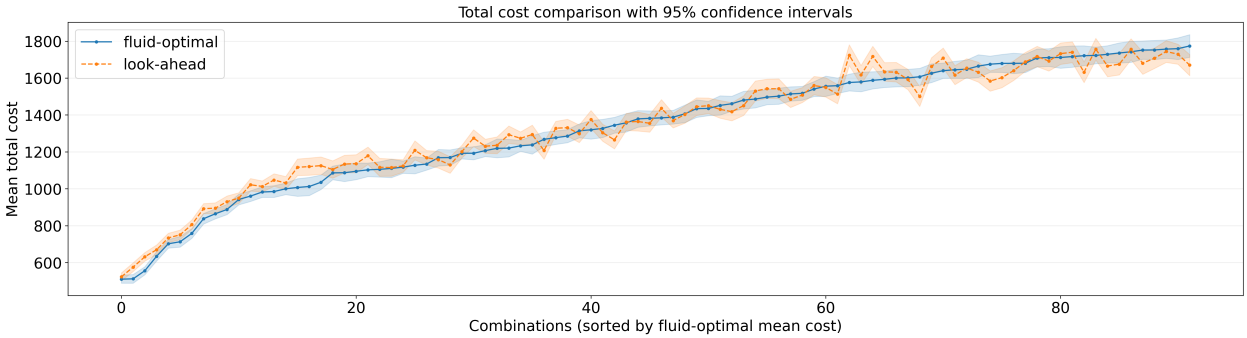


Figure 14 Two-class look-ahead policy and fluid-optimal benchmark.